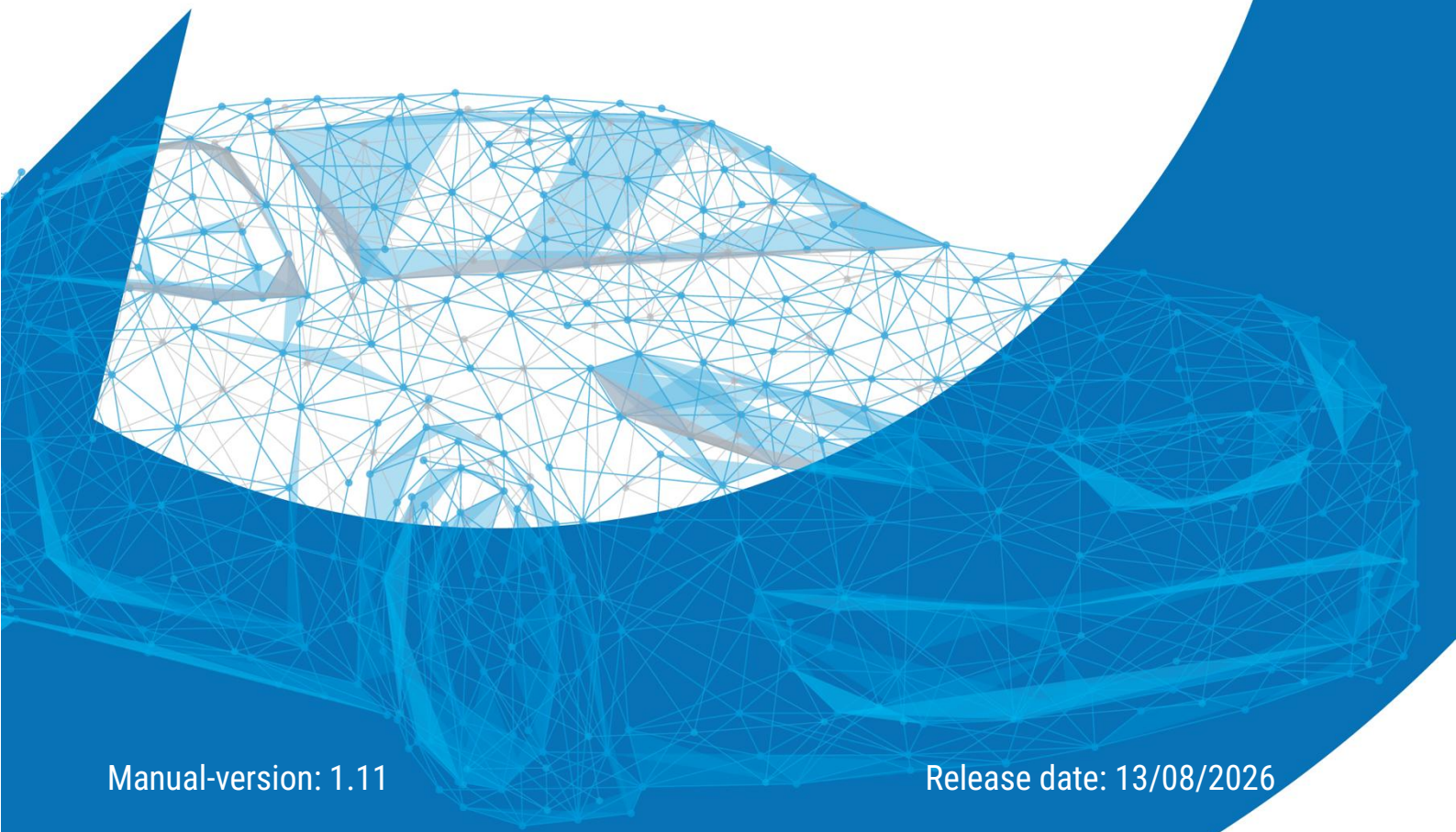
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#OneStepAhead

Enhanced Ethernet Switch

User Manual



Manual-version: 1.11

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1 Introduction

If you have any questions or need further clarification, please contact our customer support at Support

1.1 Validity

This user manual is valid for the variants:

Product Version	Software Version
v3.4	v017.004

Table 1-1: User manual validity

Ensure that the User Manual refers to the hardware variant you are using, and the firmware version installed on it.

The User Manual is valid from the time of publishing to the publication of a new version of this User Manual.

Current version in the Customer portal
The firmware is constantly being further developed, and the User Manual is therefore updated accordingly. The latest version of the User Manual can contain different or additional information or instructions. For this reason, always download the latest version of the User Manual from the Technica Engineering´s Customer Portal from https://portal.technica-engineering.de/downloads, before using the device refer to the instructions which are presented in the User Manual.

1.2 Important notes

We provide our important notes and safety instructions in two languages: Englisch (EN) and German (DE). For more details, see the relevant sections:

- EN: Important notes: Important notes
- DE: Wichtige Hinweise: Wichtige Hinweise

1.3 Warranty

We reserve the right to update the contents of this documentation and software without prior notice to reflect improvements or changes. We welcome feedback to enhance our products in the future.

The information in this manual is protected by copyright laws. Unauthorized reproduction or distribution of any part of this manual is prohibited and subject to legal action. For permission or further details, please contact the Technica Engineering directly.

1.4 Glossary

Term	Definition / Explanation
°C	degrees in Celsius
API	Application Programming Interface
approx	approximately

AVB	Audio Video Bridging
AVTP	(Layer 2) Audio-Video Transport Protocol
BASE-T	Physical layer of standard ethernet
BASE-T1	Physical layer of automotive ethernet
Ch	Channel
Com	Communication
Config	Configuration
CPU	Central Processing Unit
CRC	Cyclic Redundancy Check
DUT	Device Under Test
E2E	End-to-end
EES	Enhanced Ethernet Switch
Eth	Ethernet
FCS	Frame Check Sequence. Error-detecting code is added at the end of an Ethernet frame
GB	Gigabit
Gbps	Gigabits per second
GND	Ground (electricity)
gPTP	Generic Precision Time Protocol, IEEE 802.1AS
HW	Hardware
ID	identifier (usually a hexadecimal-number)
IEEE	Institute of Electrical and Electronics Engineers
IFG	The interframe gap between two ethernet frames
IP	Internet Protocol (Ethernet)

IP	Ingress Protection class (HW related)
IT	Information Technology
IVN	In-Vehicle Network
IVN2ETH	In-Vehicle Network to Ethernet
LED	Light Emitting Diode
MAC	Media Access Control
MACsec	network security standard that operates at the MAC layer (IEEE 802.1AE)
Mbps	Megabits per second
MKA	MACsec Key Agreement
mm	milli meter
MQS	Tyco Electronics Micro Quad Lock System, specific shape and size for crimps
MTU	maximum transmission unit
ns	Nanoseconds
OS	Operating system
OSI model	Open Systems Interconnection model
PTP	Precision Time Protocol, IEEE 1588, IEEE 802.1AS
PTPv2	Precision Time Protocol version 2, IEEE 1588-2008
RJ45	Registered Jack 45, most common connector for wired networks
SW	Software
Sync	Synchronization
Sys	System
Tab	Tabulator
TECMP	Technically Enhanced Capture Module Protocol, used for encapsulation of captured in-vehicle network data

TSN	Time Sensitive Networking
UINT	Unsigned integer
VLAN	Virtual Local Area Network
WUP	Wake Up

2 Product description

Ethernet is a point-to-point technology. This means to construct a network based on Ethernet, a switch is needed. The purpose of the switch is to intelligently communicate between the devices connected to it avoiding dedicated point-to-point connections.

Technica Engineering's Enhanced Ethernet Switch (EES) allows users to connect and loop through up to 8 10/100/1000BASE-T-based devices. This switch can establish virtual point-to-point connections using single VLAN tags (802.1Q) and double VLAN tags (802.1q-in-q), thus enabling filtering and control over data streams (mirroring and forwarding). This platform allows strong AVB/TSN capabilities intended for testing and analyzing onboard vehicle networks.

2.1 Order information

Product Name	Article Number	Cable set art. number*	Advanced license package**	MACsec feature package***
Enhanced Ethernet Switch MACsec MATEnet	TE-2003	KS-2003	TE-2003-AF	TE-2003_MACsec
Enhanced Ethernet Switch MACsec H-MTD	TE-2004	KS-2004	TE-2004-AF	TE-2004_MACsec

*Cable set needs to be ordered separately
**Advanced license package is described in CHAPTER 4.4.4
***MACsec license described in CHAPTER 4.4.5

2.2 General description

The main task of the Technica Engineering Enhanced Ethernet Switch is to act as a switch for the data collected from in-vehicle (IVN) communication systems via its multiple ports. Other features such as time synchronization of time-aware systems, segmentation of traffic, and supervising various streams of traffic are added intelligence provided by the switch. In this way, a user can set up specific solutions fitting the user's needs. The device is equipped with standard ethernet interfaces making it possible to use a regular computer for configuration and analysis.



Figure 2-1: Enhanced Ethernet Switch MACsec

2.3 Features

- Automotive MACsec and MKA capability
- Easy configuration via webserver and remote configuration messages
- Import and export of configurations
- Wake-up / sleep functionality
- Possibility to reset to default settings by push button
- Time synchronization support for 802.1AS-2011 (gPTP) and partially supporting IEEE 1588-2008 (PTPv2) and 802.1AS-2020
- Traffic Shaping (802.1Qav)
- Individual use cases for the following TSN functionalities can be provided on-demand Qbv, Qbu (check with our Technical Sales Team)
- Robust steel case
- Advanced Filtering
- Remote API
- Ingress Rate Limiter

2.4 Feature Packages

For the EES MACsec MATEnet and EES MACsec H-MTD there is currently one advanced feature package as well as one MACsec feature package available. The features can be enabled by purchasing the corresponding feature package.

2.4.1 Core Feature Package

When purchasing an EES MACsec MATEnet or EES MACsec H-MTD the core feature package is provided by default.

- Single Partition Time Synchronization
- Advanced Filter
- Remote API

2.4.2 Advanced Feature Package

The Advanced feature package includes the following features:

- Multi Partition Time Synchronization
- PTPv2togPTP Bridge
- Traffic Shaping
- Ingress Rate Limiter

2.4.3 MACsec Feature Package

The MACsec feature package includes the following features:

- MACsec + MKA

2.5 Technical data

Electrical, thermal and mechanical specifications of Product name:

Operating Temperature	-40 °C to +85 °C
Supply Voltage	6.5 V to 32 V DC (typ. 12 V)
Power consumption	6 to 13 Watt
IP Protection Class	IP 20
Housing Dimensions	198 mm (W), 141 mm (L), 39 mm (H)
Weight	0.7 kg (approx.)
Interfaces	8x 100/1000BASE-T1 (MATEnet/H-MTD) 2x SFP+ 2x PPS (In/Out) 1x 100BASE-TX (Host Port) 1x Service Port 2x Wake-Up Line

Operating temperature is the guaranteed range of temperature under which all the device features are available and working. Outside this range, the user operates the device at their own risk.

The temperature conditions where the device can be placed while not in use should not exceed 50° for the device to have consistent performance for longer periods.

When SFP+ interfaces are used then the operating temperature can be *a maximum of up to 60°C* as the SFP modules generate extra heat.

2.5.1 Power Consumption

It consumes power according to the number of ports in service. In the normal operation mode when no 100/1000BASE-T1 ports are connected, at 12 volts, the devices consume 450 mA. With each connection, the current consumption is raised by 20 mA. If all eight ports are connected to the switch, the current consumed would be approximately 610 mA.

When an SFP module is inserted into the switch, the amount of current consumed depends on the type of SFP module used. A power level 3 module (worst case scenario) consumes up to 2 W (250 mA approximately). In the worst-case scenario, the power consumed by a fully loaded EES is 450 mA + 160 mA + 500 mA = 1100 mA.

2.6 Startup Time

The following table shows the startup time values for the Enhanced Ethernet switch.

PowerUp and Processor Boot Time: The time from power-up until all hardware related to switching operations is configured	80 ms
100BASE-T1 LinkUp Time (worst case)	90 ms
1000BASE-T1 LinkUp Time (worst case)	80 ms

Table 2-1:Startup time

The 100BASE-TX Host port needs about 3-4 seconds of linkup time in Auto negotiation mode.

3 Hardware Interfaces

The Enhanced Ethernet Switch has been designed for multiple use cases. Therefore, several hardware versions of the switch have been developed so far. Differences between the versions are collected in the table below:

Product Version	Properties
v1.0	With MATEnet connectors for 1000BASE-T1 interfaces (article number TE-2003) With H-MTD connectors for 1000BASE-T1 interfaces (article number TE-2004)
v2.0	Internal changes
v3.0	Internal optimization
v3.1	Minor changes
v3.2	Minor changes
v3.3	Minor improvements
V3.4	Components sourcing update due to obsoletes.

Table 3-1: Product version history

3.1 Connectors and controls

The device is equipped with several connectors and control elements that are distributed between the front and back sides of the device. This document focuses on

- The MATEnet variant
- The H-MTD variant

There is a separate user manual for:

- Enhanced Ethernet Switch MACsec Hybrid variant
- Enhanced Ethernet Switch RJ-45 variant

The front side of the user can find:

- 1x Power connector including
  - 2x Wake up lines
  - 1x CAN-FD connection
  - 1x ToD serial port (not available for current SW status)
- 1x RJ-45 Ethernet connector (for configuration only)
- 1x Coax connector for PPS In signal (not available for current SW status)
- 1x Coax connector for PPS Out signal
- 2x SFP+ slots
- Several LEDs
- 1x Service port

The backside of the device is reserved for the interfaces used to 100/1000BASE-T1 devices. This means:

- 8x 100/1000BASE-T1 ports

3.1.1 Connectors

Here users can see an overview of all the HW interfaces of the Enhanced Ethernet Switch



Figure 3-1:Front side of Enhanced Ethernet Switch

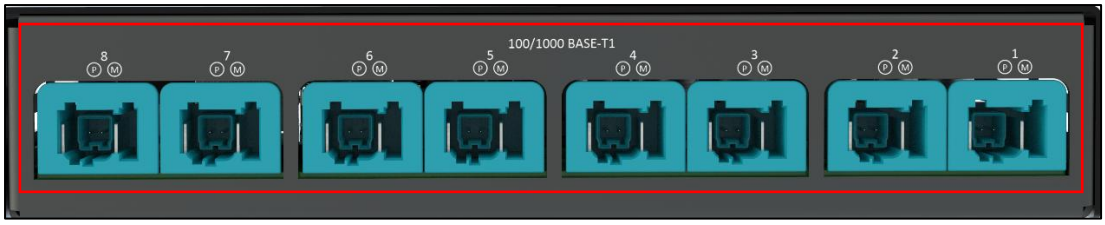


Figure 3-2 Backside of MATEnet

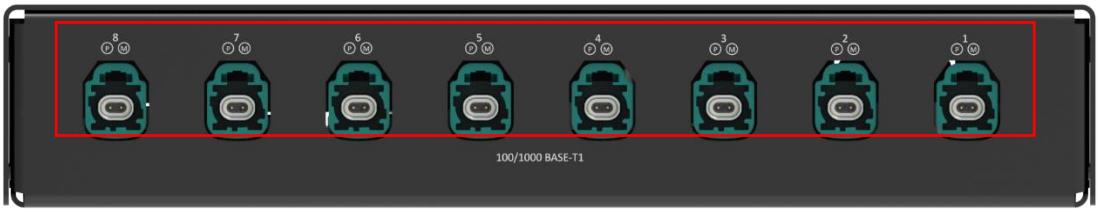


Figure 3-3: Backside of H-MTD

3.1.2 Push-button (1)

The push-button will reset all settings to factory default settings. Press this button for 5 seconds. After that:

- IP Address is set to default (10.0.0.200)
- The complete configuration is lost and is set to default

If the device is blocked and all LEDs are lit (not blinking), the device must be sent to Technica Engineering. To avoid problems please make sure that the Host LED (orange) is blinking slowly (once per second) and the power supply is stable before the reset.

3.1.3 Power connector (2)

A black MQS connector on the front side (12 pins, two rows) is used to supply the power to the Enhanced Ethernet Switch.

- The power supply for the device is supplied by pins 7 and 8 (12 Volt) and pins 1 and 2 (Ground)
- Requirements for the device itself: 12 Volt DC up to 1.1 ampere (typically 450 mA).

Next to the pins for the power supply, the pins corresponding to two independent wake-up lines can be found, identified with the letter “W” in the device housing. For each of them:

- As a digital input, it is used to wake our device or stay awake. The wake-in voltage range is from 6 V to 30 V, and the line consumes approximately 0.7 mA for the range of 6 V to 15 V and 16.5 mA for the range of 15 V to 30 V.
- As digital output, it is used to wake up external devices or keep them awake. When using the wake output feature, the Enhanced Ethernet Switch device input supply (VBAT) range is limited from 9 V to 16 V. The wake-out line is able to supply a maximum of 25 mA for the EES input supply (VBAT) range of 9 V to 12 V or up to 60 mA for EES input supply range of 12 V to 16 V.

For information about wake-up functionality please check CHAPTER 4.3.5.

Next to the pins for wake-up lines, the pins for a CAN-FD connection can be found. These pins are not yet supported by the software.

Pins 11 and 12 are for Time of the Day (ToD). It is an RS-232 port that is intended to connect an external GPS to the EES. This function is not yet supported by the software.

The connector Tyco Electronics Micro Quad Lock System (MQS) is used, as described in the table below:

Name	Picture	Part Number
TE connectivity MQS cover 12P		1-1534096-1
MQS terminals housing 12P		1534100-1
TE MQS crimp contact		928999-1

Table 3-2: Parts of Tyco MQS connector

Pin	Function	Pin	Function
1	Power Ground	7	Battery +12 Volt Input
2	Power Ground	8	Battery +12 Volt Input
3	WakeUP 1	9	Not connected
4	WakeUP 2	10	Not connected
5	CAN-FD High	11	ToD Serial Port TX
6	CAN-FD Low	12	ToD Serial Port RX

Table 3-3: Pinning of black MQS connector

3.1.4 Coax connector for PPS In Signal (4)

An SMA coaxial connector is used to input a One Pulse Per Second (PPS) signal to synchronize with any external device providing this signal. The function associated with this connector is currently not supported by the software.

3.1.5 Coax connector for PPS Out Signal (4)

An SMA coaxial connector is used to output a One Pulse Per Second (PPS) signal that allows an external device to synchronize with EES.

3.1.6 SFP slots (5)

There are two SFP slots for SFP+ modules in the switch. For details of configuration, please check CHAPTER 4.1.5

The following modules have been tested and can be used with the switch:

- BEL SFP-1GBT-05 SFP 10/100/1000 Mbps RJ-45 Module.
- 10Gtek 10GBASE-T SFP+ Transceiver, RJ-45 SFP+ CAT.6a up to 30 meters.

Technica Engineering SFP Modules are also compatible with the switch. For further information about Technica Engineering SFP units please contact Technical Sales.

3.1.7 Host port (6)

There is one RJ-45 Standard Ethernet connector for Fast Ethernet on the front side, labeled as 100BASE-TX Host, that can be used for the configuration of the device.

The in-built LEDs in the RJ-45 connector show the status of the 100BASE-TX configuration port.

- The left (red LED) is lit by a link-up.
- The right (orange LED) will blink on data traffic.

3.1.8 Service connector (7)

The Enhanced Ethernet Switch is provided with an interface to be used for service only.

3.1.9 MATEnet connector (8)

The MATEnet variant is equipped with 8x 100/1000BASE-T1 ports allowed through the MATEnet connectors.

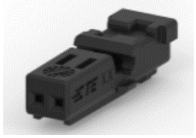
Name	Picture	Part Number
1-PORT MATENET FRAME, UTP, COD Z		2302454-9
Tyco MATENET, FEMALE INSERT MODUL, BLACK		2302450-1
Tyco Nano MQS crimp contact		1-1703930-2

Table 3-4: Parts of Tyco MATEnet connector

Users can use the official Tyco tool for these crimp contacts. But we recommend purchasing a ready-to-use cord set.

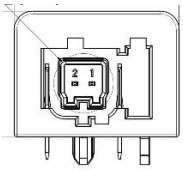


Figure 3-4: MATEnet connector

Pin	Function
1	Data Line Minus (Negative)
2	Data Line Plus (Positive)

Table 3-5: MATEnet connector

3.1.10 H-MTD connector (9)

The H-MTD variant is equipped with 8x 100/1000BASE-T1 ports allowed through the H-MTD connectors

Name	Picture	Part Number
Rosenberger H-MTD Connector		E6K10A-1CAZ5-Z

Table 3-6: Parts of Rosenberger H-MTD connector

Pinning:



Figure 3-5: H-MTD connector

Pin	Function
1	Data Line Plus (Positive)
2	Data Line Minus (Negative)

Table 3-7: H-MTD connector

3.1.11 Other interfaces

These interfaces include LED indicators for various interfaces.

3.1.11.1 Status LEDs (3)

LED	Description
Host LED	Green: Always lit, during normal operation of the switch Orange: Blink slowly (once/second) during normal operation of the device Blink fast (flicker) suggesting the device is in boot-loader mode
Switch LED	Green; Currently not in use Orange; Currently not in use
SFP	The 2 SFP LEDs show if the inserted SFPs in the SFP+ slots have a linkup

Table 3-8: Details for Status LEDs

3.1.11.2 Link LEDs (4)

There are three types of link LEDs corresponding to the interface

1. The 8 port status LEDs of all 100/1000BASE-T1 ports monitor the link status of the corresponding port.
   - LED permanently on indicates there is LinkUp in the corresponding port.
   - LED blinking means communication on the corresponding port.

Whenever P/M Pins of the 100/1000BASE-T1 Link are swapped, an auto-correction of plus/minus is possible.

2. The 2 SFP LEDs show if the inserted SFPs in the SFP+ slots have a linkup
3. The in-built LEDs in the RJ-45 connector show the status of the 100BASE-T1 configuration port.
   - The right (orange LED) Permanently “on” indicates there is LinkUp, blinking means communication on the port.

4 Configuration

The Enhanced Ethernet Switch is a highly configurable device, offering the user great flexibility to include it in many different scenarios.

Out-of-the-box, access to the webserver is possible by connecting the device via the Host port (RJ-45) to a computer. The default IP address for the device is 10.0.0.200. The recommended subnet mask is 255.255.255.0.

Once connected to the device (through the 100BASE-TX Host RJ-45 connector):

1. Please go to the network adapter settings and adjust the IP in the same range as your Enhanced Ethernet Switch. For example, 10.0.0.20 with subnet mask 255.255.255.0.
2. Enter the IP address in your standard web browser. Verified browsers include Microsoft Edge, Google Chrome, and Mozilla Firefox.

4.1 Home screen

The following website should appear. The user can select the corresponding ports for further configuration.

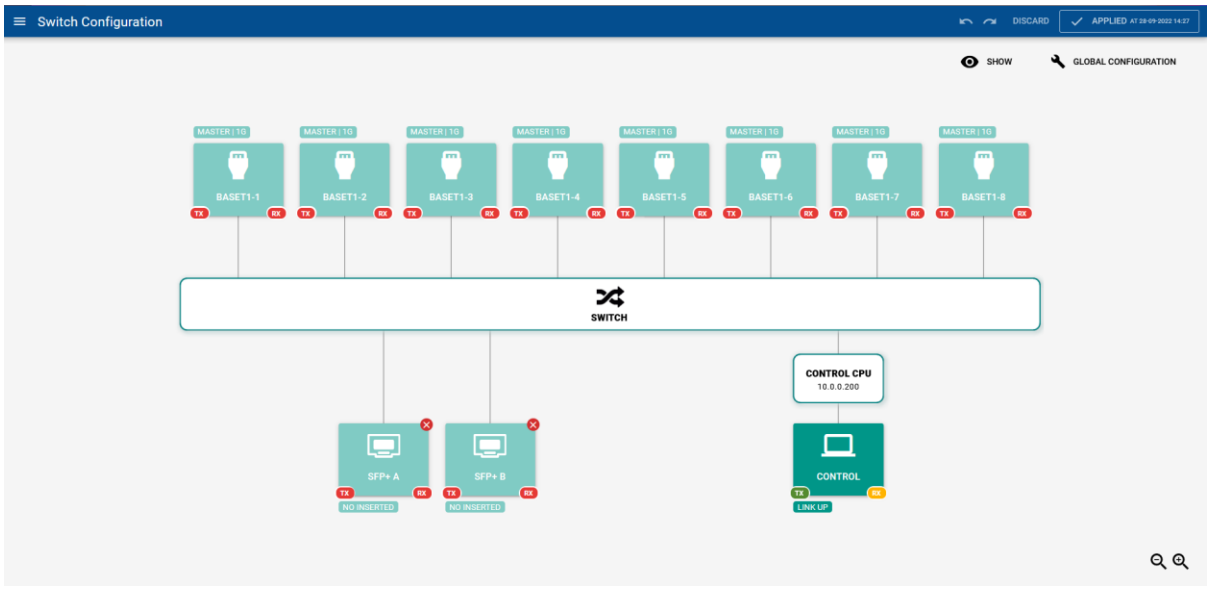


Figure 4-1: Home Screen

Please select one of the tabs for further configuration. Each tab is described in detail in the next sections of this document.

Any change in the configuration will lead to the following hint appearing at the top of the current web page:





This option sends data from the webpage to the memory of the device. There is no need to save every time this hint appears. All the desired changes can be done at once and the configuration can be saved at the end. To do that, please click *"SAVE CONFIGURATION"*. Please note that the configuration will not be applied unless the device is restarted.



By clicking on this option, the user is prompted to introduce an optional name for the configuration file and to accept the changes. By selecting *"OK"* the configuration is saved similar to the option *"SAVE CONFIGURATION"*. It also restarts the device, applying the saved configuration.

Note that without *"SAVE & APPLY CONFIGURATION"*, the configuration of the device will not be fully updated. To be sure that all changes have been applied, a restart of the device is necessary.

4.1.1 Switch configuration tab

The following options can be explored in the switch configuration tab.

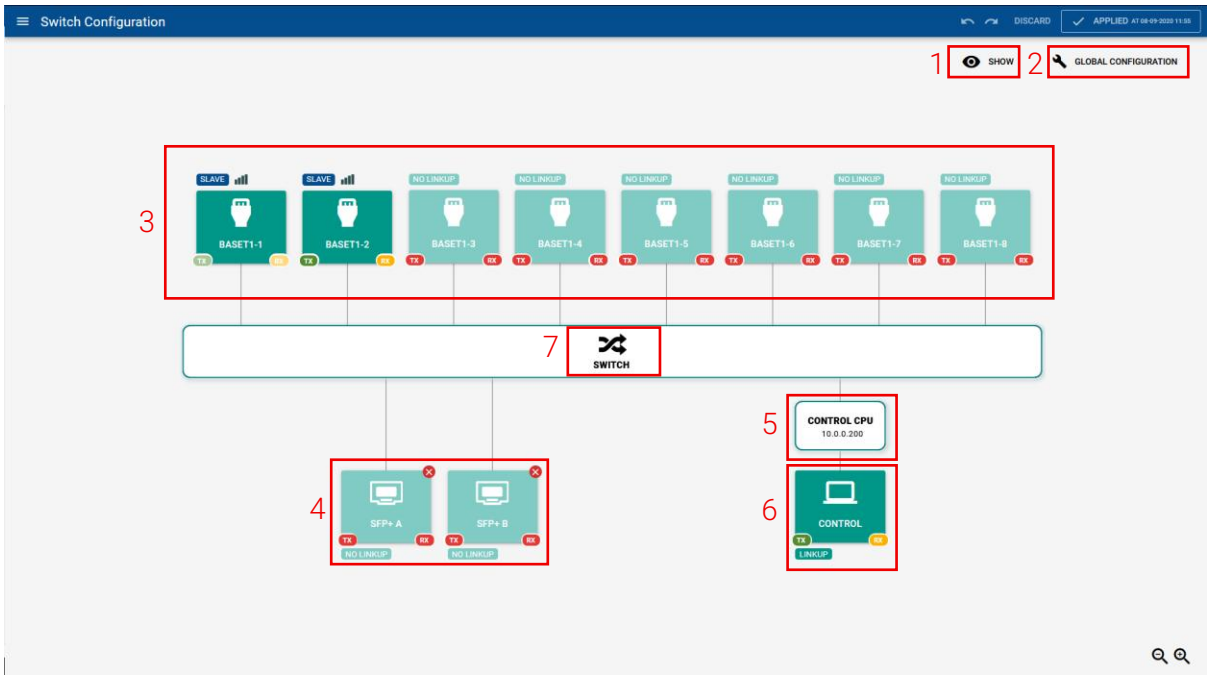


Figure 4-2: Switch Configuration Tab

4.1.2 Show configuration (1)

This tab visualizes the settings on the switch

- *Show VLANs*: When this option is checked, the user can add VLAN IDs that can be visualized on the switch configuration page. A maximum of 7 sections are allowed

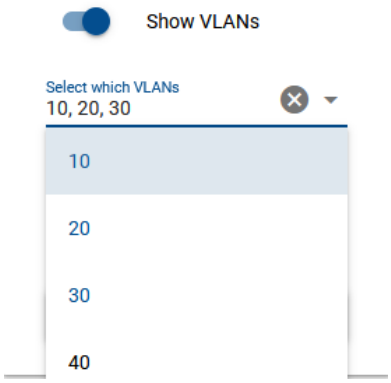


Figure 4-3: Selection of VLANs

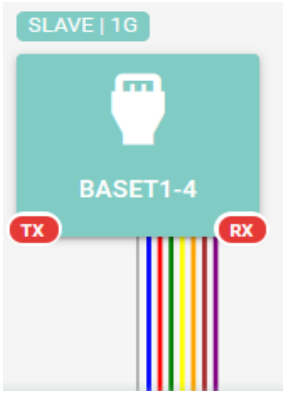


Figure 4-4: Visualization of VLANs

- *Show Mirroring:* When this option is checked, all the mirroring ports are visualized. Mirroring can only be selected on the SFP+ A and SFP+ B ports. When visualized mirroring is shown as "dotted lines."

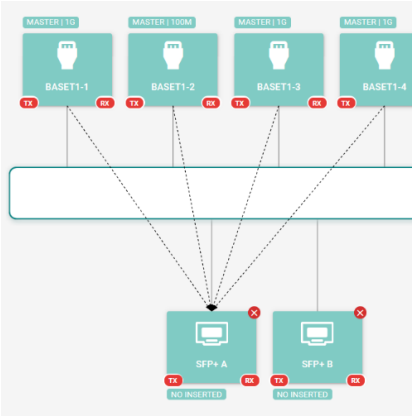


Figure 4-5: Visualization of Mirroring

- *Show Port Segmentation:* When this option is checked, all the configured segments are visualized.

4.1.3 Global configuration (2)

This option permits the user to enable port settings on the switch.

- *Port Segmentation:* When this option is checked, the user can make segments of ports that are allowed to exchange traffic. CHAPTER 4.1.8.1.
- *VLAN:* When this option is checked, the user can configure VLAN Tags (802.1q) and is allowed to facilitate VLAN-based traffic. See CHAPTER 4.1.8.3
- *Double Tagging:* When this option is checked, the user can configure double VLAN Tags (802.1q-in-q) and facilitate double VLAN-tagged traffic. The user is also free to enter any TPID for the second tag in hex. By default, the TPID is 0x88A8. See CHAPTER 4.1.8.3
- *Time Sync:* When this option is checked, the user can configure the ports of the switch for gPTP. The user can create partitions for time synchronization. See CHAPTER 4.1.8.6.
- *Advanced Filter:* When this option is checked, the user can define rules for filtering traffic. See CHAPTER 4.1.8.4.
- *Traffic Shaping:* When this option is checked, the user can configure 802.1Qav related parameters on each port. See CHAPTER 4.1.8.5.
- *Ingress Rate Limiter:* When this option is checked, the user can configure token buckets for streams of traffic on ingress. See CHAPTER 4.1.8.7
- *Address Table:* Here the user can view and manipulate entries on the address resolution table. See CHAPTER 4.1.8.2

ⓘ At a time, only one option between *VLAN* and *Double Tagging* can be selected.

4.1.4 BASE-T1 ports (3)

The switch is provided with 8 BASE-T1 ports (BASE-T1-1 to BASE-T1-8). The switch configuration tabs show basic information about the configuration of the port. This includes:

- *MASTER:* The selected BASE-T1 port is Master, and the connected DUT should be in Slave mode
- *SLAVE:* The selected BASE-T1 port is Slave, and the connected DUT should be in Master mode
- *100M/1G:* The selected BASE-T1 port is configured at 100Mbps/1Gbps
- *Signal Strength:* The select port has the following signal strength for TX/RX
- *Automatic BASE-T1 parameters search:* A loupe icon on the top left indicates that the automatic BASE-T1 search is enabled, and role and speed will be detected automatically.
- *Reverse Polarity:* A warning icon in the top right shows that the positive and negative wires in the connection have been swapped.



Master role at 1Gbps | *Slave role at 100Mbps* | *Automatic Parameters Search* | *Cable polarity inversion detection*

Figure 4-6: BASEt1 Ports Visualization

4.1.4.1 Port summary

The user can see the summary of the configuration of a port along with some statistics by placing the cursor over the desired port.

4.1.4.2 Port configuration

The user can click on the desired port for configuration. The configuration parameters are distributed between general configuration and specific configuration for MACsec.

General configuration

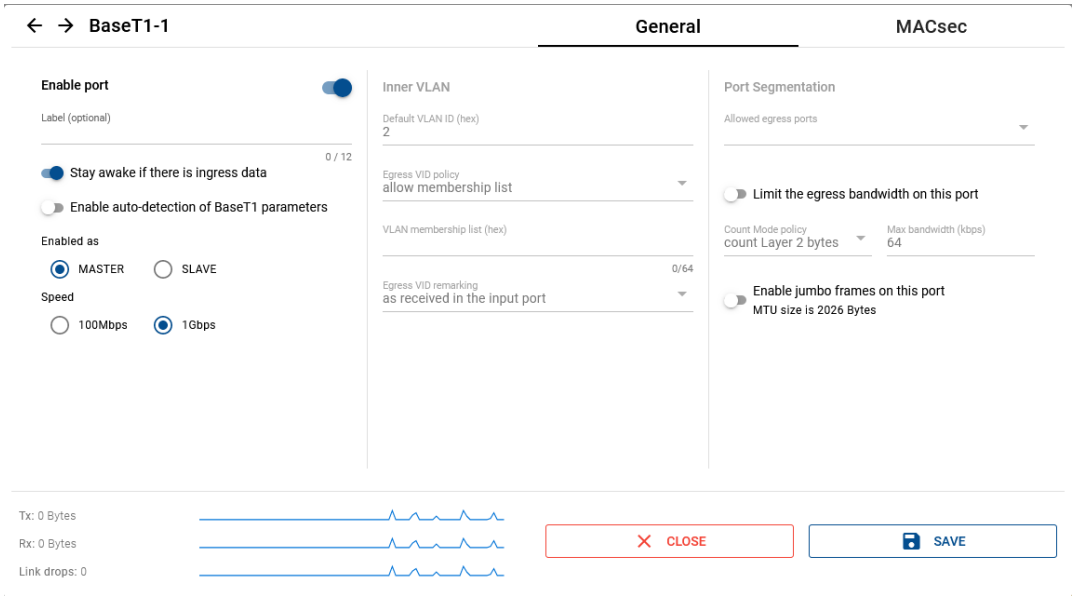


Figure 4-7: General configuration menu for BASET1-1Port

Enable Port: The user can enable or disable the port. Once the port is disabled the color of the port is changed to grey.

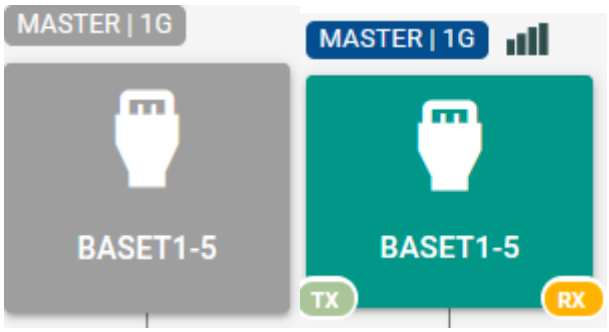


Figure 4-8: Disabled and Enabled Port

Label: The user can add a name for a particular port. A maximum of 12 characters can be used for the name.

Stay awake: When this option is enabled the Port stays up as long as there is traffic on the port.

Auto-detection of BaseT1 parameters: When this option is enabled the device tries all combinations to identify the speed (100 or 1000 Mbps) and the role (master or slave). The role (master/slave) and speed options are greyed out when this option is enabled, as the equipment will detect the neighbouring device's configuration by itself.

Enabled as:

- *MASTER:* The selected BASE-T1 port is Master, and the connected DUT should be in Slave mode
- *SLAVE:* The selected BASE-T1 port is Slave, and the connected DUT should be in Master mode

Speed:

- *100Mbps:* The selected BASE-T1 port is operating with 100Mbps speed
- *1Gbps:* The selected BASE-T1 port is operating with 1Gbps speed

TX: The decimal value of sent octets on this port along with the graph

RX: The decimal value of received octets on this port along with the graph

Link drops: The decimal value of the number of times the link was dropped on the connected ports along with the graph

Inner/Outer VLAN: In this section, the user can change and add settings about VLANs. Depending on the selection made in *Global Configuration*, the user can configure Inner or Outer VLANs.

ⓘ The switch cannot be enabled to configure inner and outer VLAN tags at the same time. So, at one point in time, only one of the two settings between *VLAN (inner VLAN)* and *Double Tagging (outer VLAN)* can be selected.

Default VLAN ID: The user can define one hexadecimal value between 0x2 and 0xFFF. Incoming traffic is single-tagged with this value, if not already tagged. If the traffic comes with a tag, the VLAN ID is not overwritten.

ⓘ VLAN ID 0x01 is used internally so it cannot be used.

Ingress and Egress VLAN policy: The user can select the VLAN policies applied for this port on ingress and egress. The user can select between several combinations for ingress and egress VLAN policy composed from the following options:

- *Allow all*: Allow all the traffic

- *Allow membership list:* Allow only the traffic that is tagged with VLANs present in the membership list
- *Allow all but exclusion list:* Allow all traffic except the ones that are tagged with VLANs present in the exclusion list

VLAN membership list: The user can input any values between 0x2 and 0xFFF. These values can be defined separated by "," [E.g.: 100, 101, EDF, …]. A range of VLANs can be defined using the character "-". [E.g. 100-105]. Outgoing traffic is allowed to egress the switch on this port if the data is tagged with one of the defined VLAN-IDs

Egress VID remarking:

- *As received in the input port:* The tags on VLAN IDs mentioned in the VLAN membership list are kept as received on the ingress
- *Removed:* The tags on VLAN IDs mentioned in the VLAN membership list are removed on the egress
- *Normalized:* If the packet is received without a VLAN tag, at egress, it is tagged with the default VLAN in the ingress port of the packet. If it is received with a VLAN tag, act as the option, *as received in the input port*.
- *Custom:* The VLANs mentioned in *the VLAN egress removed list* are removed while the VLANs mentioned in *the VLAN egress normalized list* are forced.

Port Segmentation: The section allows the user to add ports to create segments within the switch. The traffic of one port segment is completely isolated from other segments.

Selected Ports: The user can mention ports here that are desired to be part of the segments. The user can select which ports the packets ingressed on this port can be forwarded to the allowed egress ports. More information can be found in the chapter on *Port segmentation*.

Limit the egress bandwidth on this port: This feature allows the user to limit the overall output bandwidth of the selected port. Traffic shaping can be set up to limit the bandwidth using the payload of layers 1 to 3 (OSI Model Layers). The following three options can be selected for traffic shaping.

- *Count Layer 1 bytes:* When selected the user can limit traffic starting from the first byte of a frame (from intergap to CRC).
- *Count Layer 2 bytes:* When selected the user can limit traffic starting after the header of the Data link layer (Layer 2) of a frame (from Destination MAC to end of payload).
- *Count Layer 3 bytes:* When selected the user can limit traffic starting after the header of the network layer (Layer 3) of a packet (only the ethernet payload).
- *Maximum bandwidth (kbps):* The user can define the bandwidth to limit in kbps. The range of this field is between 64kbps to 1000000kbps. More information can be found in the chapter for bandwidth reservation.

Enable jumbo frames on this port: This feature allows the user to enable or disable the reception of jumbo frames on the selected port.

- When the option is disabled the maximum transmission unit (MTU) is 2048 Bytes for VLAN tagged frames and 2044 for untagged frames.
- When the option is enabled the maximum transmission unit (MTU) is 10240 Bytes for VLAN tagged frames and 10236 for untagged frames.



When Jumbo frames are enabled all time-aware functions of the switch are affected. This involves features like *Time Sync, Ingress Rate Limiter, Traffic Shaping,* etc.

To identify any possible error in the cabling, the EES reports if a cable polarity inversion is detected. If the port is set to 100 Mbps polarity inversion can only be identified if the PHY is set as slave. If the port is set to 1Gbps polarity inversion can be identified regardless of whether the role is master or slave. Once polarity inversion is detected the EES shows a sign in the top right corner of the port.

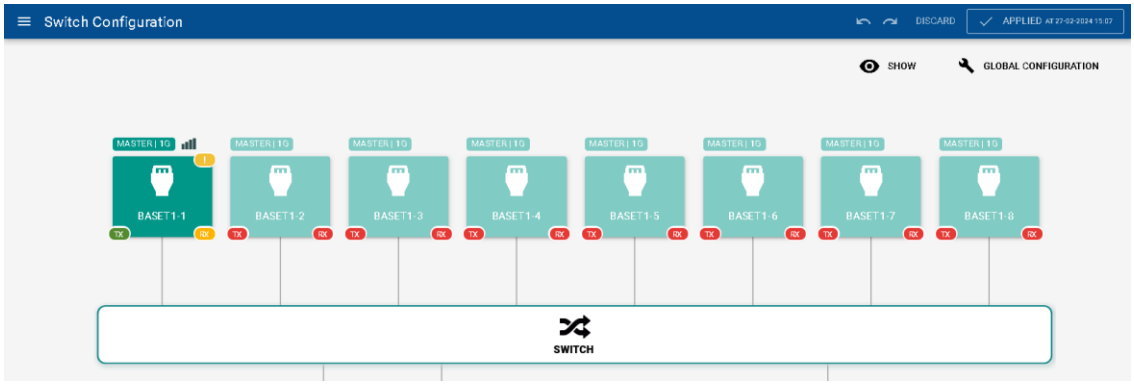


Figure 4-9: Polarity inversion detected

MACsec configuration

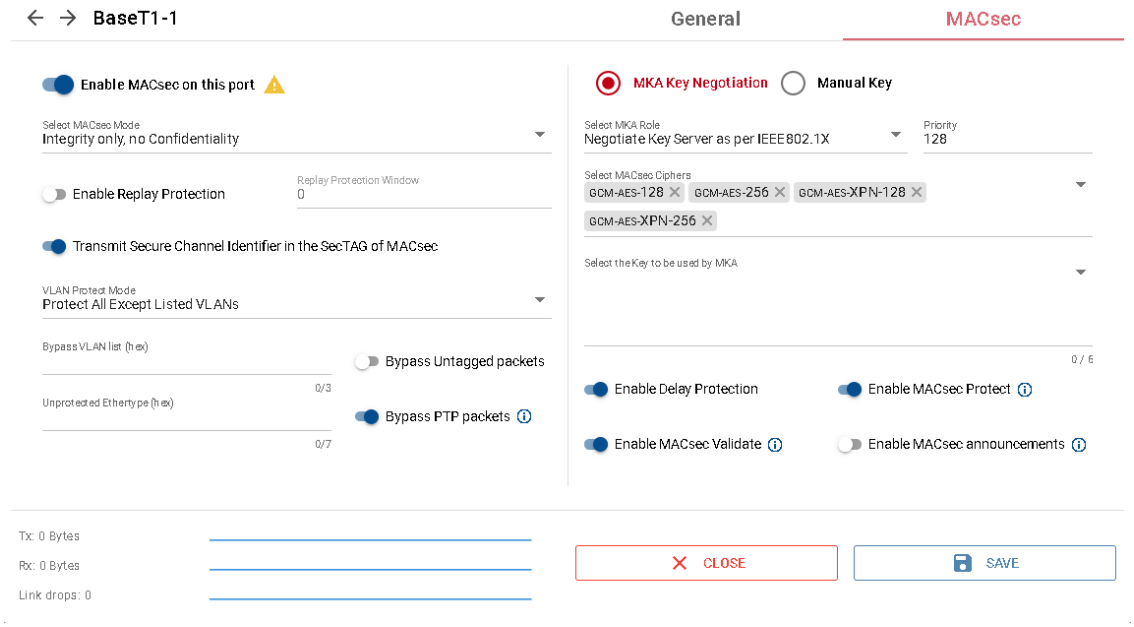


Figure 4-10: MACsec configuration menu for BASET1-1 Port

Enable MACsec on this port: This option enables/disables the MACsec functionality on this port. By default, the option is disabled.

Select MACsec mode: The user can select possible settings for MACsec mode, from the below selections.

- *Neither integrity nor Confidentiality*: MKA negotiation without MACSEC
- *Integrity only, no Confidentiality*: The payload is protected with an integrity signature, but the data is not encrypted
- *Integrity and Confidentiality, offset 0*: The payload is protected with an integrity signature, and all the payload is encrypted
- *Integrity and Confidentiality, offset 30:* The payload is protected with an integrity signature, and all the payload is encrypted except the first 30 bytes
- *Integrity and Confidentiality, offset 50*: The payload is protected with an integrity signature, and all the payload is encrypted except the first 50 bytes

By default, the value is set to “integrity only, no confidentiality”.

Enable Replay Protection: This option enables/disables replay protection functionality on this port. By default, the option is disabled.

Replay protection window: This field enables the user to specify the replay protection window size. It's only applied if enable replay protection is active. By default, the value is 0. Possible values are all uint32 values.

Transmit secure channel identifier in the SecTAG of MACsec: This option implies that the MACsec tag will contain the MAC and port of the entity with whom it negotiated. Given that this is a switch, by default, the option is enabled.

VLAN protect mode: This option sets which VLANs need to be protected. The user can select one of the two options:

- Protect all except listed VLANs
- Protect only listed VLANs

By default, the option **Protect all except listed VLANs** is selected.

Bypass VLAN/ Unprotected VLAN (hex): This field gives a list of VLANs to protect or bypass. By default, the field is empty. Possible values are from 1 to 4095. Up to 3 values can be added in protect mode while 2 values can be added in Bypass mode.

Unprotected EtherType (hex): This field gives a list of EtherTypes that are not going to be encrypted (up to 7 EtherTypes can be added to this list). By default, the field is empty.

Bypass PTP packets: If this field is enabled and PTP is configured for this port, PTP messages will not be protected by MACsec. By default, the option is enabled.

MKA Key negotiation: When selected the MKA key exchange negotiation is enabled and MKA node configuration is used. If this option is not selected, then the port is forced to install the TX and RX keys specified in the **Manual Key**.

MKA Key negotiation parameters: Role for the principal actor, possible settings:

Negotiate key server as per IEEE802.1X: Dynamically elect key server as per IEEE802.1X
Act as a key server, no role negotiation: Act as the key server, no negotiation
Act as a key client, no role negotiation: Act as a key client, no negotiation

By default, the value is set to "Negotiate key server as per IEEE802.1X". Possible values are only the integer values that are defined by MKA enumeration.

Priority: This value is applicable when the MKA role is configured as "Negotiate key server as per IEEE802.1X. By default, the value is 128. The possible range of values is between 0 to 255. The device with the highest priority (lowest value) will play the role of the server. When the role is forced to the key server, the value is internally set to the highest priority (0). Similarly, when the role is forced to a key client, the value is internally set to the lowest priority (255).

Select MACsec ciphers: This option provides a list of ciphers in order of preference. A non-listed cipher is not permitted. Possible ciphers values are:

- GCM_AES_128
- GCM_AES_256
- GCM_AES_XPN_128
- GCM_AES_XPN_256

Select the key to be used by MKA: This option allows the user to select which key the MKA will use from the keys defined in the global MACsec/MKA configuration.

Enable delay protection: When enabled, it performs frequent updates of the packet number on the receiving side to prevent the possibility of an attacker delaying MACSEC frames. By default, the option is disabled.

Enable MACsec Protect: Protect outgoing frames (see IEEE 802.1X CP FSM – "macsecProtect"). By default, the option is enabled.

Enable MACsec Validate: Drop invalid incoming frames (see IEEE 802.1X CP FSM – "macsecValidate"). By default, the option is enabled.

Enable MACsec Announcements: Enables MACsec Cipher Suites TLV in MKPDU Announcement parameter (secure announcements). By default, the option is disabled.

Disable cipher suite negotiation: Disables the automatic negotiation of the cipher suite. By default, the option is enabled.

Manual Key: When selected the MKA key has to be manually configured by the user. If this option is not selected, then the other choice for the user is *MKA Key Negotiation*.

Manual Key Parameters: The user can select the cipher to be used in case of manual configuration. Possible choices are:

- GCM_AES_128
- GCM_AES_256
- GCM_AES_XPN_128
- GCM_AES_XPN_256

TX Config: There user must configure the TX parameters here.

TX Side Association Number (AN): The user must input the association number. By default, the value is 0. Possible values are between 0 to 3

TX Side Packet Number (PN): The user must input the packet number used by the cipher. By default, the value is 1. Possible values are all uint64_t values except 0 if XPN cipher is used, otherwise, all 32bit values are used except 0.

TX Slide Secure Association Key (SAK): The user must provide the SAK value. The possible values are between 0, 16 bytes, or 32 bytes. Any size other than these values will result in an error.

TX Side Secure Channel Identifier (SCI): The user must provide the value of Secure Channel Identifier value. The possible values are between 0 and 8 bytes. Any size other than these values will result in an error.

TX Side Secure Channel Identifier (SCCI): This field stores an unsigned integer of 32 bits. This can only be used in XPN ciphers. By default, the value is 0. There is no restriction on possible values.

TX Side Salt: This field stores 12 bytes as a value for the Salt used in XPN ciphers. By default, the value is kept empty. Possible values are between 0 and 12 bytes. Any size other than these values will result in an error.

RX Config: There user must configure the TX parameters here.

RX Side Association Number (AN): The user must input the association number. By default, the value is 0. Possible values are between 0 to 3

RX Side Packet Number (PN): The user must input the packet number used by the cipher. By default, the value is 1. Possible values are all uint64_t values except 0 if XPN cipher is used, otherwise, all 32bit values are used except 0.

RX Slide Secure Association Key (SAK): The user must provide the SAK value. The possible values are between 0, 16 bytes, or 32 bytes. Any size other than these values will result in an error.

RX Side Secure Channel Identifier (SCI): The user must provide the value of Secure Channel Identifier value. The possible values are between 0 and 8 bytes. Any size other than these values will result in an error.

RX Side Secure Channel Identifier (SCCI): This field stores an unsigned integer of 32 bits. This can only be used in XPN ciphers. By default, the value is 0. There is no restriction on possible values.

RX Side Salt: This field stores 12 bytes as a value for the Salt used in XPN ciphers. By default, the value is kept empty. Possible values are between 0 and 12 bytes. Any size other than these values will result in an error.

Copy Manual Key to: This allows the user to copy a manually configured MACsec key to other MACsec capable ports.

4.1.5 SFP+ Ports (4)

The switch is provided with 2 SFP+ ports that are capable of supporting up to 10Gbps standards 10GBASE-R and USXGMII. Users can insert 10G capable SFP modules in these slots to operate these ports in 10Gbps mode.

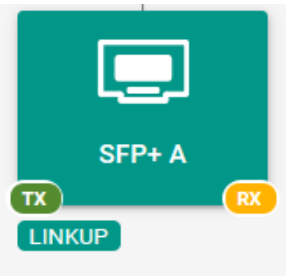


Figure 4-11: SFP+ Port Visualization

4.1.5.1 Port Summary

The user can see the summary of the configuration of a port along with some statistics by placing the cursor over the desired port.

4.1.5.2 Port configuration

The user can click on the desired port for configuration. The configuration parameters are distributed between general configuration and specific configuration for MACsec.

General configuration

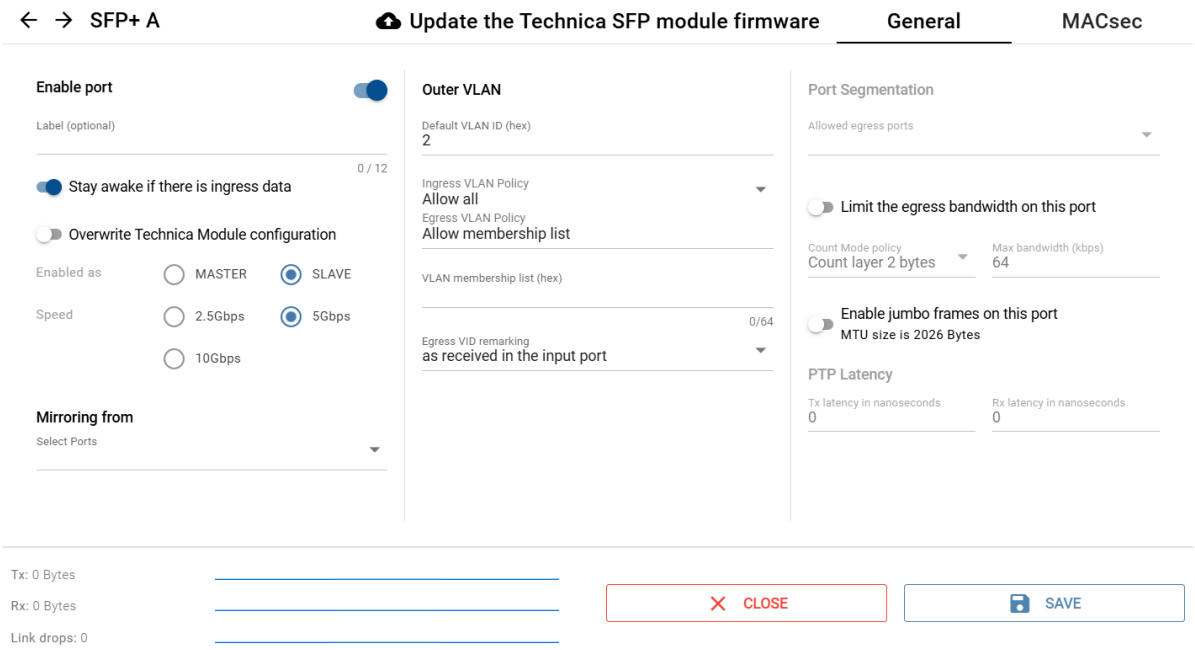


Figure 4-12: General configuration menu for SFP+ A Port

Enable Port: The user can enable or disable the port. Once the port is disabled the color of the port is changed to grey.

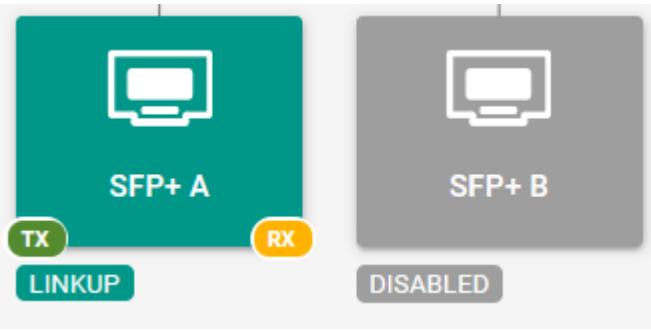


Figure 4-13: Disabled and Enabled Port

Label: The user can add a name for a particular port. A maximum of 12 characters can be used for the name.

Stay awake: When this option is enabled the Port stays up as long as there is traffic on the port

The subsequent information varies based on the inserted SFP module

Generic SFP Module	Technica SFP Module
Mode: This mode sets the speed on the SERDES Bus between the EES and the SFP module. It is not setting the speed between the SFP module and the connected device on the outer side. • *SGMII:* This option needs to be selected when the SFP cages are equipped with 100/1000BASE-T1 SFPs. Some of the 1G SFP Modules are not working with this option, then please check the 1000BASE-X option. • *10GBaseR:* This option needs to be selected when the SFP cages are equipped with 10G SFPs, also when the connection between connected NIC and the SFP module is using a different speed. • *Other modes:* Other modes are 1000BASE-X, 2500BASE-X, 5GBASE-R and USXGMII. The 1G,2.5G and 5G modes should be used, when the maximum speed of the SFP module is also 1G, 2.5 or 5G. USCGMII is the same as SGMII for 10G Modules, but this standard is not often used in SFP modules.	Technica SFP Modules are equipped with a DIP switch for configuration. For details please refer to the respective SFP User Manual. *Overwrite Technica Module configuration:* This option allows the user to overwrite the DIP switch configuration of the SFP module via Software. When enabled the following options can bet set: • *Enabled as*: This allows the user to configure between Master and Slave. • *Speed:* This allows the user to configure the speed of the SFP port.

Mirroring from: The following option can be used to mirror the traffic on the selected ports to the selected SFP+ port. All the traffic that is received on the ports added in the selected ports panel will be mirrored on the given SFP+ port.

TX: The decimal value of sent octets on this port along with the graph.

RX: The decimal value of received octets on this port along with the graph.

Link drops: The decimal value of the number of times the link was dropped on the connected ports along with the graph.

Inner/Outer VLAN: In this section, the user can change and add settings about VLANs. Depending on the selection made in *Global Configuration*, the user can configure Inner or Outer VLANs.

The switch cannot be enabled to configure inner and outer VLAN tags at the same time. So, at one point in time, only one of the two settings between *VLANs (inner VLAN)* and *Double Tagging (outer VLAN)* can be selected.

Default VLAN ID: The user can define one hexadecimal value between 0x2 and 0xFFF. Incoming traffic is single-tagged with this value, if not already tagged. If the traffic comes with a tag, the VLAN ID is not overwritten

VLAN ID 0x01 is used internally so it currently cannot be used

Ingress and Egress VLAN policy: The user can select the VLAN policies applied for this port on ingress and egress. The user can select between several combinations for ingress and egress VLAN policy composed from the following options:

- *Allow all:* Allow all the traffic to be egressed by this port
- *Allow membership list:* Allow only the traffic that is tagged with VLANs present in the membership list
- *Allow all but exclusion list:* Allow all traffic except the ones that are tagged with VLANs present in the membership list

VLAN membership list: The user can input any values between 0x2 and 0xFFF. These values can be defined separated by "," [E.g.: 100, 101, EDF, …]. A range of VLANs can be defined using the character "-". [E.g. 100-105]. Outgoing traffic is allowed to egress the switch on this port if the data is tagged with one of the defined VLAN-IDs

Egress VID remarking:

- *As received in the input port:* The tags on VLAN IDs mentioned in the VLAN membership list are kept as received on the ingress
- *Removed:* The tags on VLAN IDs mentioned in the VLAN membership list are removed on the egress
- *Normalized:* If the packet is received without a VLAN tag, at egress, it is tagged with the default VLAN in the ingress port of the packet. If it is received with a VLAN tag, act as the option, *as received in the input port.*
- *Custom:* The VLANs mentioned in *the VLAN egress removed list* are removed while the VLANs mentioned in *the VLAN egress normalized list* are forced.

Port segmentation: The section allows the user to add ports to create segments

Selected ports: The user can mention ports here that are desired to be part of the segments. The user can select which ports the packets ingressed on this port can be forwarded to the allowed egress ports. More information can be found in the chapter on Port segmentation.

Limit the egress bandwidth on this port: This feature allows the user to limit the overall output bandwidth of the selected port. Traffic shaping can be set up to limit the bandwidth using the payload of layers 1 to 3 (OSI Model Layers). The following three options can be selected for traffic shaping.

- *Count Layer 1 bytes:* When selected the user can limit traffic starting from the first byte of a frame (from intergap to CRC).
- *Count Layer 2 bytes:* When selected the user can limit traffic starting after the header of the Data link layer (Layer 2) of a frame (from Destination MAC to end of payload).
- *Count Layer 3 bytes:* When selected the user can limit traffic starting after the header of the Network layer (Layer 3) of a packet (only the Ethernet payload).
- *Maximum bandwidth (kbps):* The user can define the bandwidth to limit in kbps. The range of this field is between 64kbps to 1000000kbps. More information can be found in the chapter for bandwidth reservation.

Enable jumbo frames on this port: This feature allows the user to enable or disable the reception of jumbo frames on the selected port.

- When the option is disabled the maximum transmission unit (MTU) is 2026 Bytes
- When the option is enabled the maximum transmission unit (MTU) is 10240 Bytes



When Jumbo frames are enabled all time-aware functions of the switch are affected. This involves features like *Time Sync, Ingress Rate Limiter, Traffic Shaping,* etc.

PTP Latency: This is an expert option for the user to compensate PTP latencies of the used SFP modules. These latencies have to be provided by the SFP manufacturer.

MACsec configuration

The EES supports MACsec and MKA configuration for Multigigabit Technica SFP modules (TE-1445) with Product Version 3.0 or later with System Release R.001.010 or newer.

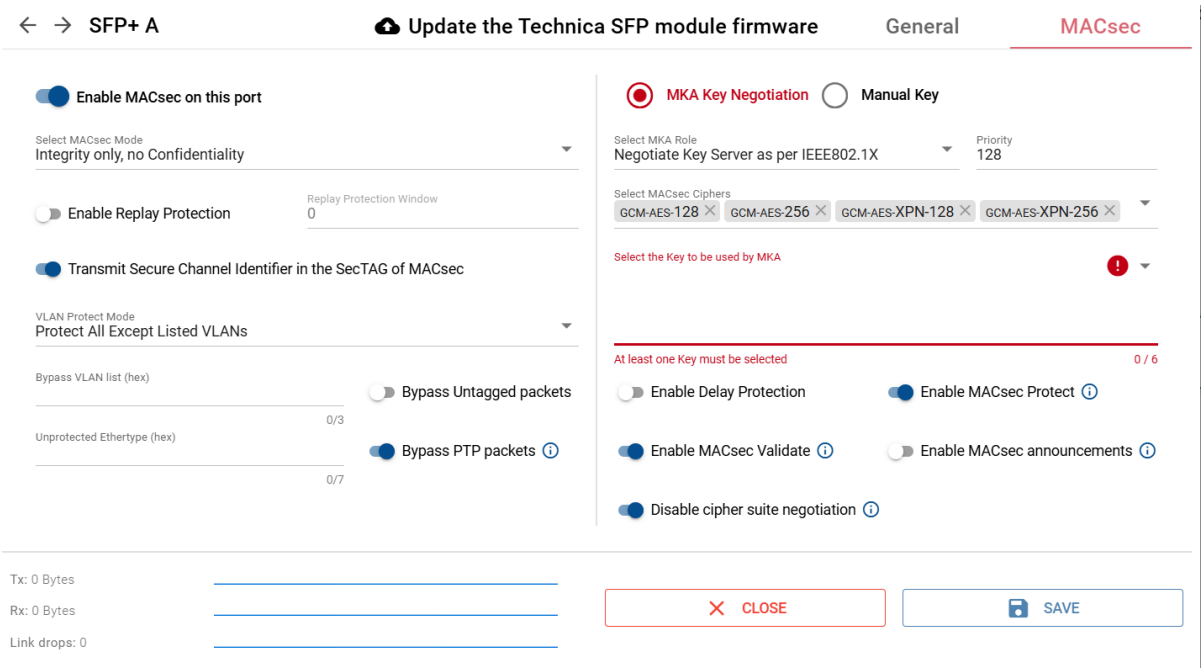


Figure 4-14: MACsec configuration menu for BASET1-1 Port

Enable MACsec on this port: This option enables/disables the MACsec functionality on this port. By default, the option is disabled.

Select MACsec mode: The user can select possible settings for MACsec mode, from the below selections.

- *Neither integrity nor Confidentiality*: MKA negotiation without MACSEC
- *Integrity only, no Confidentiality*: The payload is protected with an integrity signature, but the data is not encrypted
- *Integrity and Confidentiality, offset 0:* The payload is protected with an integrity signature, and all the payload is encrypted
- *Integrity and Confidentiality, offset 30:* The payload is protected with an integrity signature, and all the payload is encrypted except the first 30 bytes
- *Integrity and Confidentiality, offset 50*: The payload is protected with an integrity signature, and all the payload is encrypted except the first 50 bytes

By default, the value is set to “integrity only, no confidentiality”.

Enable Replay Protection: This option enables/disables replay protection functionality on this port. By default, the option is disabled.

Replay protection window: This field enables the user to specify the replay protection window size. It’s only applied if enable replay protection is active. By default, the value is 0. Possible values are all uint32 values

Transmit secure channel identifier in the SecTAG of MACsec: This option implies that the MACsec tag will contain the MAC and port of the entity with whom it negotiated. Given that this is a switch, by default, the option is enabled.

VLAN protect mode: This option sets which VLANs need to be protected. The user can select one of the two options:

- Protect all except listed VLANs
- Protect only listed VLANs

By default, the option Protect all except listed VLANs is selected.

Bypass VLAN/ Unprotected VLAN (hex): This field gives a list of VLANs to protect or bypass. By default, the field is empty. Possible values are from 1 to 4095. Up to 3 values can be added in protect mode while 2 values can be added in Bypass mode.

Unprotected EtherType (hex): This field gives a list of EtherTypes that are not going to be encrypted (up to 7 EtherTypes can be added to this list). By default, the field is empty.

Bypass PTP packets: If this field is enabled and PTP is configured for this port, PTP messages will not be protected by MACsec. By default, the option is enabled.

MKA Key negotiation: When selected the MKA key exchange negotiation is enabled and MKA node configuration is used. If this option is not selected, then the port is forced to install the TX and RX keys specified in the Manual Key.

MKA Key negotiation parameters: Role for the principal actor, possible settings:

Negotiate key server as per IEEE802.1X: Dynamically elect key server as per IEEE802.1X
Act as a key server, no role negotiation: Act as the key server, no negotiation
Act as a key client, no role negotiation: Act as a key client, no negotiation

By default, the value is set to "Negotiate key server as per IEEE802.1X". Possible values are only the integer values that are defined by MKA enumeration.

Priority: This value is applicable when the MKA role is configured as "Negotiate key server as per IEEE802.1X. By default, the value is 128. The possible range of values is between 0 to 255. The device with the highest priority (lowest value) will play the role of the server. When the role is forced to the key server, the value is internally set to the highest priority (0). Similarly, when the role is forced to a key client, the value is internally set to the lowest priority (255).

Select MACsec ciphers: This option provides a list of ciphers in order of preference. A non-listed cipher is not permitted. Possible ciphers values are:

- GCM_AES_128
- GCM_AES_256
- GCM_AES_XPN_128
- GCM_AES_XPN_256

Select the key to be used by MKA: This option allows the user to select which key the MKA will use from the keys defined in the global MACsec/MKA configuration.

Enable delay protection: When enabled, it performs frequent updates of the packet number on the receiving side to prevent the possibility of an attacker delaying MACSEC frames. By default, the option is disabled.

Enable MACsec Protect: Protect outgoing frames (see IEEE 802.1X CP FSM – "macsecProtect"). By default, the option is enabled.

Enable MACsec Validate: Drop invalid incoming frames (see IEEE 802.1X CP FSM – "macsecValidate"). By default, the option is enabled.

Enable MACsec Announcements: Enables MACsec Cipher Suites TLV in MKPDU Announcement parameter (secure announcements). By default, the option is disabled.

Disable cipher suite negotiation: Disables the automatic negotiation of the cipher suite. By default, the option is enabled.

Manual Key: When selected the MKA key has to be manually configured by the user. If this option is not selected, then the other choice for the user is *MKA Key Negotiation*.

Manual Key Parameters: The user can select the cipher to be used in case of manual configuration. Possible choices are:

- GCM_AES_128
- GCM_AES_256
- GCM_AES_XPN_128
- GCM_AES_XPN_256

TX Config: There user must configure the TX parameters here.

TX Side Association Number (AN): The user must input the association number. By default, the value is 0. Possible values are between 0 to 3

TX Side Packet Number (PN): The user must input the packet number used by the cipher. By default, the value is 1. Possible values are all uint64_t values except 0 if XPN cipher is used, otherwise, all 32bit values are used except 0.

TX Slide Secure Association Key (SAK): The user must provide the SAK value. The possible values are between 0, 16 bytes, or 32 bytes. Any size other than these values will result in an error.

TX Side Secure Channel Identifier (SCI): The user must provide the value of Secure Channel Identifier value. The possible values are between 0 and 8 bytes. Any size other than these values will result in an error.

TX Side Secure Channel Identifier (SCCI): This field stores an unsigned integer of 32 bits. This can only be used in XPN ciphers. By default, the value is 0. There is no restriction on possible values.

TX Side Salt: This field stores 12 bytes as a value for the Salt used in XPN ciphers. By default, the value is kept empty. Possible values are between 0 and 12 bytes. Any size other than these values will result in an error.

RX Config: There user must configure the TX parameters here.

RX Side Association Number (AN): The user must input the association number. By default, the value is 0. Possible values are between 0 to 3

RX Side Packet Number (PN): The user must input the packet number used by the cipher. By default, the value is 1. Possible values are all uint64_t values except 0 if XPN cipher is used, otherwise, all 32bit values are used except 0.

RX Slide Secure Association Key (SAK): The user must provide the SAK value. The possible values are between 0, 16 bytes, or 32 bytes. Any size other than these values will result in an error.

RX Side Secure Channel Identifier (SCI): The user must provide the value of Secure Channel Identifier value. The possible values are between 0 and 8 bytes. Any size other than these values will result in an error.

RX Side Secure Channel Identifier (SCCI): This field stores an unsigned integer of 32 bits. This can only be used in XPN ciphers. By default, the value is 0. There is no restriction on possible values.

RX Side Salt: This field stores 12 bytes as a value for the Salt used in XPN ciphers. By default, the value is kept empty. Possible values are between 0 and 12 bytes. Any size other than these values will result in an error.

Copy Manual Key to: This allows the user to copy a manually configured MACsec key to other MACsec capable ports.

4.1.6 CPU port (5)

This port allows the user to change the MAC Address and the IP Address of the switch.

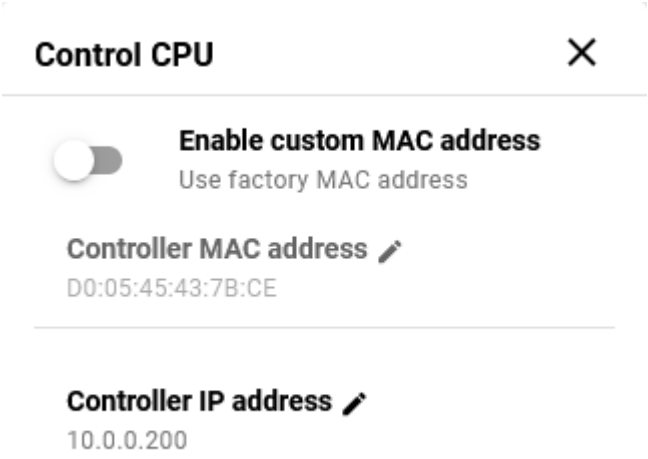


Figure 4-15: Control CPU

Technica assigns a unique MAC address to each device. If this MAC address needs to be changed, the custom MAC address toggle must be enabled and then the Controller MAC Address field must be populated with the new MAC address. The factory MAC address is never lost, so disabling the custom MAC address toggle will force the device to ignore the controller MAC address entered and use the factory MAC address.

By default, the IP address is 10.0.0.200.

The MAC and IP Address can also be changed in the *NETWORK DETAILS* (See CHAPTER 4.3.6.1 section of the *SYSTEM INFORMATION TAB*).

4.1.7 Control port (6)



Figure 4-16: Enabled Control Port

The user can click on the desired port for configuration.

4.1.7.1 Port summary

The user can see the summary of the configuration of a port along with some statistics by placing the cursor over the desired port.

4.1.7.2 Port configuration

The user can click on the desired port for configuration.

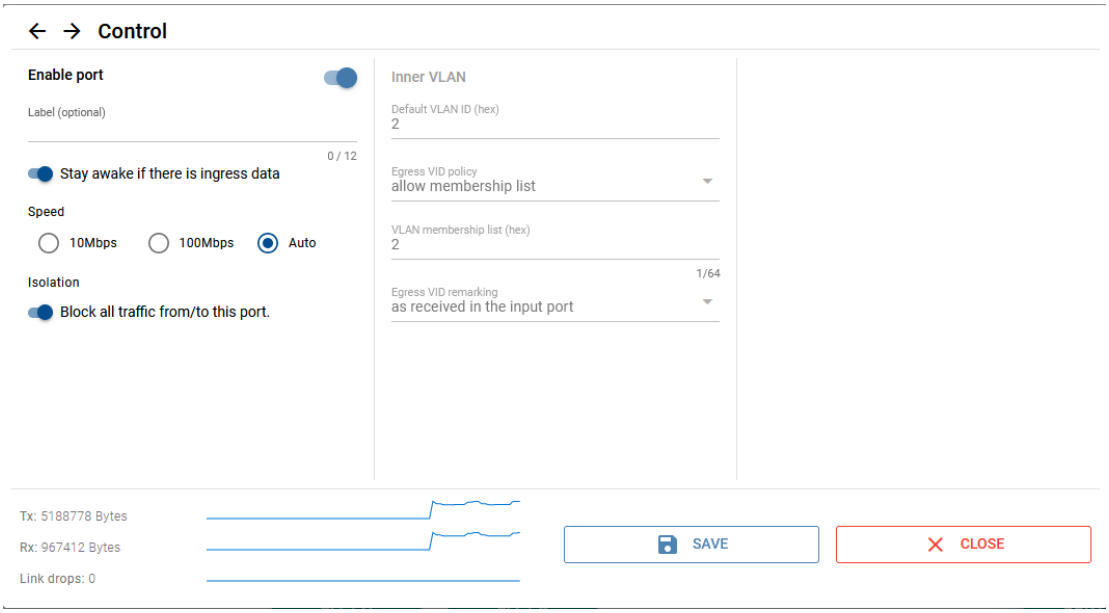


Figure 4-17: Configuration Menu for Control Port

Enable Port: This port cannot be disabled because this port is used for the configuration of the device:

Label: The user can add a name for a particular port. A maximum of 12 characters can be used for the name.

Stay awake: When this option is enabled the Port stays up if there is traffic on the port.

Speed:

- *10Mbps:* The selected 100BASE-TX configuration port is operating with 10Mbps speed
- *100Mbps:* The selected 100BASE-TX configuration port is operating with 10Mbps speed
- *Auto*: The selected 100BASE-TX configuration port is operating in auto-negotiation mode

Isolation: This option can be used to isolate traffic to or from the configuration port to the rest of the ports of the switch

TX: The decimal value of sent octets on this port along with the graph.

RX: The decimal value of received octets on this port along with the graph.

Link drops: The decimal value of the number of times the link was dropped on the connected ports along with the graph.

Inner/Outer VLAN: In this section, the user can change and add settings about VLANs. Depending on the selection made in *Global Configuration*, the user can configure Inner or Outer VLANs.

The switch cannot be enabled to configure inner and outer VLAN tags at the same time. So, at one point in time, only one of the two settings between *IEEE 802.1q mode (inner VLAN)* and *Double Tagging (outer VLAN)* can be selected.
Default VLAN ID: The user can define one hexadecimal value between 0x2 and 0xFFF. Incoming traffic is single-tagged with this value, if not already tagged. If the traffic comes with a tag, the VLAN ID is not overwritten.

VLAN ID 0x01 is used internally so it currently cannot be used.

Ingress and Egress VLAN policy: The user can select the VLAN policies applied for this port on ingress and egress. The user can select between several combinations for ingress and egress VLAN policy composed from the following options:

- *Allow all:* Allow all the traffic to be egressed by this port
- *Allow membership list:* Allow only the traffic that is tagged with VLANs present in the membership list
- *Allow all but exclusion list:* Allow all traffic except the ones that are tagged with VLANs present in the membership list

VLAN membership list: The user can input any values between 0x2 and 0xFFF. These values can be defined separated by "," [E.g.: 100, 101, EDF, …]. A range of VLANs can be defined using the character "-". [E.g. 100-105]. Outgoing traffic is allowed to egress the switch on this port if the data is tagged with one of the defined VLAN-IDs

Egress VID remarking:

- *As received in the input port:* The tags on VLAN IDs mentioned in the VLAN membership list are kept as received on the ingress
- *Removed:* The tags on VLAN IDs mentioned in the VLAN membership list are removed on the egress
- *Normalized:* If the packet is received without a VLAN tag, at egress, it is tagged with the default VLAN in the ingress port of the packet. If it is received with a VLAN tag, act as the option, *as received in the input port*.
- *Custom:* The VLANs mentioned in *the VLAN egress removed list* are removed while the VLANs mentioned in *the VLAN egress normalized list* are forced.

4.1.8 Switch

This tab is used to configure the switch. The various functions offered by the Enhanced Ethernet Switch can be configured and modified graphically using this tab.

4.1.8.1 Port segmentation

This sub-tab is used to isolate the subsets of ports. When this option is enabled in *Global configuration* (See CHAPTER 4.1.3), the user can create subsets where the user can allow the traffic ingressed to be egressed only out of the mentioned ports. This way traffic on some ports can be isolated from other ports. For example, in Figure 4-17 a port segment is created between BASET1-4 and BASET1-5. The traffic ingressed on BASET1-4 can only be egressed to BASET1-5. In case bi-directional isolation is needed BASET1-5 must enable BASET1-4, that way the traffic ingressed on BASET1-5 can be egressed on BASET1-4 as well as establishing a bi-directional isolation tunnel for the traffic exchanged on these two ports. The user must connect both ports but if "Edit Symmetrically" is enabled, the device automatically selects the other way when a connection is enabled.
The following options can be found in this tab:

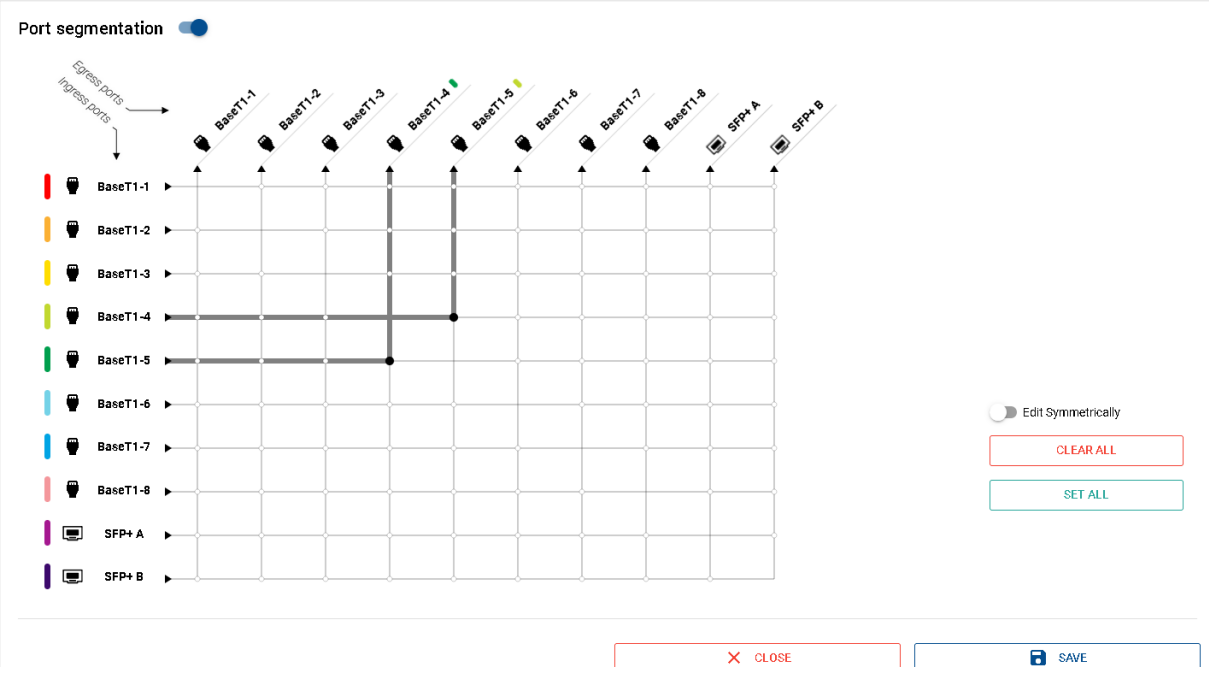
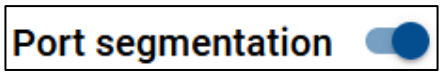
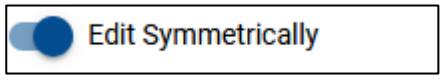


Figure 4-18: Port segmentation



Enables the feature port segmentation



Enables bi-directional communication between port segments. Disabling this option creates only one-way port segments



Deletes all the configured port segments



Create port segments between all ports

Port segmentation can work alongside VLANs. However, it is not possible to override port segmentation by VLAN configuration. Port segmentation has more priority over VLAN configuration.

4.1.8.2 Adress resolution table

This sub-tab is used to display the mapping of MAC addresses to the physical ports.

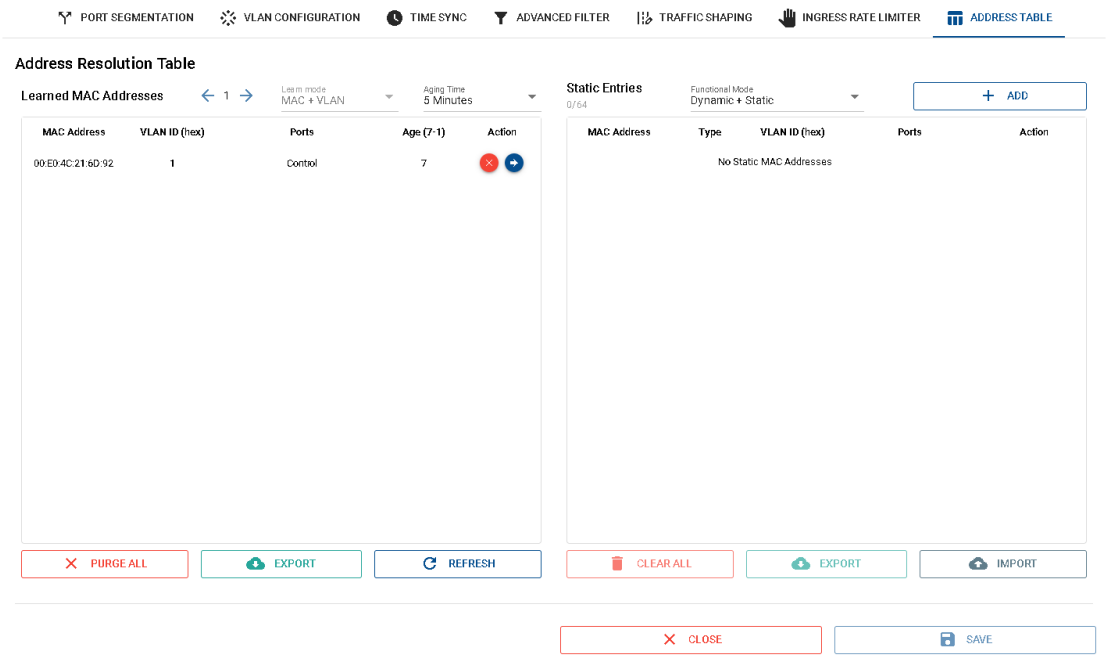


Figure 4-19: Address Resolution Table

The left-hand side shows a table of *Learned MAC Addresses*. When the VLAN mode is enabled, the user has the choice how the *Learn mode* will learn the MAC Addresses for the table. Either the MAC Address will be learned together with the VLAN, or the table can learn only the MAC Address. The user can select the *aging time* to specify for how long the switch wants to remember the corresponding MAC address. If the aging time is "disabled" the switch never forgets all learned MAC addresses. The following buttons can be used in the section of *Learned MAC Addresses*.

Opens a dialog to create a new static MAC entry using the learned MAC address information

Deletes the corresponding entry

PURGE ALL — Deletes all the entries in the Learned MAC Address table

EXPORT — The learned table can be exported to the connected PC

REFRESH — Refreshes the table for new entries or removes expired entries

The right-hand side shows a table of manually added Static (and Dynamic) entries. The user can select the functional mode of the address resolution table to *static+dynamic* or only static entries. Up to 64 static entries can be added by the user in this table. To add the entries the user can select and input the MAC address and port manually.

ADD

The following buttons can be used in the section of *Static Entries*.

Enables the user to create or edit a new entry for mapping between the MAC address and the port

Enables the user to clone an existing entry for mapping between the MAC address and the port

Deletes the corresponding entry

CLEAR ALL — Deletes all the entries in the Learned MAC Address table

EXPORT — The learned table can be exported to the connected PC

IMPORT — The learned table can be imported from the connected PC
The tables can be imported or exported in .json format

While adding an entry in the addressing resolution table, ingress VLANs can be used to distinguish packets from different endpoints that have the same MAC addresses but different VLANs.

4.1.8.3 VLAN configuration

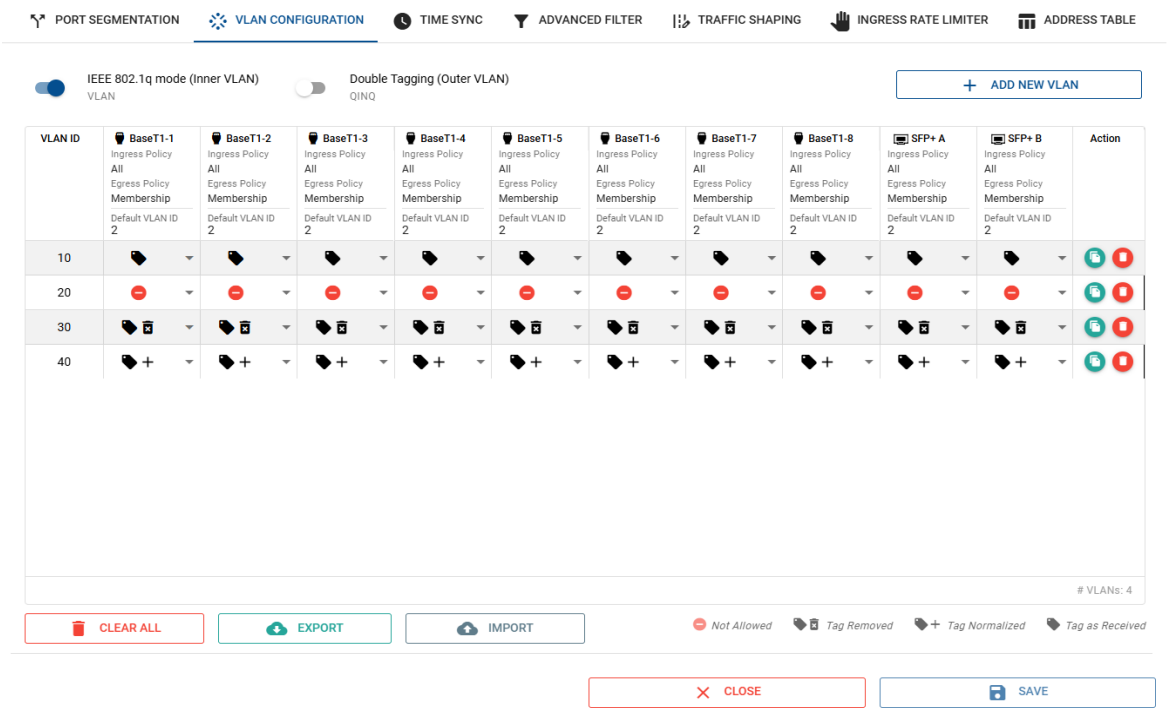


Figure 4-20: Configuration of VLANs

If *IEEE 802.1q mode (inner VLAN)* is enabled port based single VLANs are activated.
In this mode, the forwarding of frames through the switch is defined by the configuration on each port. Users can also make these configurations using this sub-tab. This sub-tab shows all the ports with the ability to different processing mechanisms based on the selection made for egress. The sections of this sub-tab are described below.

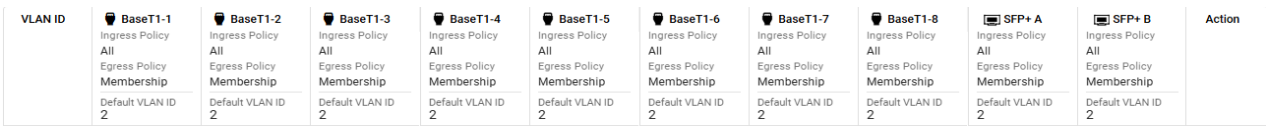


Figure 4-21: Egress VID Policy

The default combination of Ingress Policy and Egress Policy is set to Allow all / Allow Membership List, so the EES behaves the same as in releases prior to 016.000

The options mentioned in Figure 4-17 provide the same parameters to be configured as in the individual port configuration for VLANs under the field *Ingress and Egress VLAN policy* (See CHAPTER 4.1.7.2). The user can select between the following options:

- *Allow all*: Allow all the traffic to be egressed by this port
- *Allow membership list:* Allow only the traffic that is tagged with VLANs present in the membership list
- *Allow all but exclusion list*: Allow all traffic except the ones that are tagged with VLANs present in the membership list

Not all possible combinations are available to be configured for Ingress and Egress VLAN Policy

When a new VLAN is added + ADD NEW VLAN using the user can select what should be done with the mentioned VLAN while egressing out of the corresponding port. The options mentioned in Figure 4-19: Address Resolution Tablecan also be configured inside the individual ports (See CHAPTER 4.1.4.2) under the field *Egress VID remarking*.

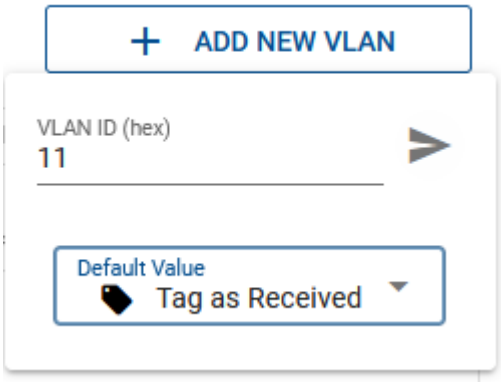


Figure 4-22: Egress VID remarking

The user can select between the following options:

- *As received in the input port:* The tags on VLAN IDs mentioned in the VLAN membership list are kept as received on the egress
- *Removed:* The tags on VLAN IDs mentioned in the VLAN membership list are removed on the egress
- *Normalized:* If the traffic is not received with an Outer VLAN then place a tag on the traffic on egress. If it is received with a tag, act as the option, as received in the input port.
- *Custom:* The VLANs mentioned in *the VLAN egress removed list* are removed while the VLANs mentioned in *the VLAN egress normalized list* are kept

These options can be edited using the panel shown in Figure 4-23: Egress VID remarking editing from one option to another.

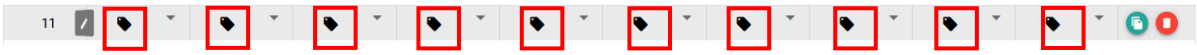


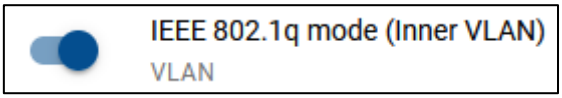
Figure 4-23: Egress VID remarking editing

The following buttons can be used in the section

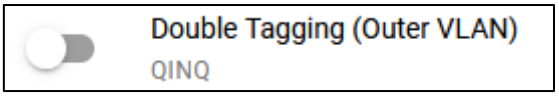
Enables the user to clone a setting (egress VID policy) for a new VLAN ID

Deletes the corresponding entry

Applies to the corresponding entry



When enabled user can configure the inner tag of VLANs (802.1q)



When enabled users can configure the outer tag of VLANs(802.1q-in-q)



Enables the user to add a new VLAN



Deletes all the configured VLANs

The switch cannot be enabled to configure inner and outer VLAN tags at the same time. So, at one point in time, only one of the two settings *VLAN (inner VLAN)* and *Double Tagging (outer VLAN)* can be selected.

4.1.8.4 Advanced filter

This sub-tab provides extended flexibility by defining filters for the received traffic. The user can define a filter rule via 

It allows the user to inspect the content of the packet. Depending on the content of the packet the user can trigger some actions.

Ingress filter match: The user must choose the ingress ports here. Specific conditions can be defined to inspect the type of traffic ingressed through these ports. Ingress filters can work in two modes.

- Normal mode
- Expert Mode

Normal mode: In normal mode, the following options are available for ingress traffic.

Is the frame double-tagged? When enabled the ingress traffic inspects the outer tag (802.1q-in-q).

Double tagging must be enabled in the *Global Configuration*

Users can select the attributes of inspection for ingress traffic. Users can also select the *AND* operator to combine more than one rule for filtering the traffic. Attributes for a protocol like Ethernet, IPv4, IPv6, and ARP can be selected. The following attributes can be used for inspection.

Destination MAC
Source MAC
Outer VLAN PCP
Outer VLAN DEI
Outer VLAN ID
Inner VLAN TPID
Inner VLAN PCP
Inner VLAN DEI
Inner VLAN ID
EtherType or Length
IP version (IPv4 / IPv6)
IPv4 Internet Header Length
IPv4 TTL
IPv4 Protocol
IPv4 Source address
IPv4 Destination address
ARP Operation
ARP Sender Hardware Address
ARP Sender Protocol Address

ARP Receiver Hardware Address
ARP Receiver Protocol Address
IPv6 Source Address
IPv6 Destination Address
IPv6 NDP Solicitation Address
IPv6 NDP Advertisement Address
IPv6 NDP Operation
IPv6 Next Header
IEEE1722 AVTP Header Type
IEEE1722 AVTP Protocol Subtype
IEEE1722 AVTP Stream ID valid
IEEE1722 AVTP Stream ID
TECMP Device ID
TECMP Message Type
TECMP Data Type
TECMP Interface ID
UDP Source Port
UDP Destination Port
TCP Source Port
TCP Destination Port

Expert mode: The user can define a filter based on any Byte that is received on the ingress port within the first 96 Bytes. The user can click on the Byte that needs to be inspected for filtering and define an *Offset*, *Data*, and *Mask*.

Two 48Bytes rules are not the same as 96 Byte rule. A total of 48 advanced filter rules can be defined.

Action: The user can define an action to be implemented on the traffic received on the ports selected in the *ingress filter match* section. The following actions can be performed.
The actions are divided into 4 subcategories:

- *Modify Port Routing:*
  (This allows us to modify the normal forwarding on the packets)
  - Mirror packets to the specified ports:
    (Copy without modifying the normal forwarding)
  - Drop packets on the specified ports:
    (Drop at egress the packet in the selected ports)
  - Forward packets only on the specified ports
    (Overwrite the forwarding ports with the configured ports)
- *Assign at Ingress:*
  (Assign some internal "remark" to the input packet)
  - Flow ID
    (Used by the Ingress Rate Limiter to mark, which traffic shall be controlled)
    (See Chapter 4.1.8.7)
  - Counter
    (Used together with the dashboard view to count how many times a rule (or set of rules) has been triggered)
- *Overwrite at Ingress*:
  (Overwrite before the forwarding engine has decide the destination ports of an ingress packet)
  - VLAN ID
    (Overwrite the VLAN ID of the ingress packet, even before the VLAN rules decide whether the packet should be accepted or dropped)
  - Priority
    (Overwrite the priority assigned by the PCP mapping, but the PCP value remains intact)

- *Overwrite at Egress:*
  (Overwrite after the forwarding engine has decide the destination port. Only the ports assigned in the “action” will apply the change)
  - VLAN Tag Remarking
    (Allows to modify at egress the VLAN remarking policy, overwrite the VLAN ID, PCP and DEI fields)

Note: VLAN mode must be enabled for Overwrite at Ingress and Egress.

If no action is selected in the advanced filter, the filter will serve to prevent further filtering of matching packets, since only one filter can be applied to each packet. This functionality can be used to exclude specific packets from a general group of packets.

For instance, to filter all IPv4 traffic except frames with Destination IP x.x.x.x, follow these steps:

1. Create an advanced filter for Destination IP x.x.x.x and select "no action."
2. Create another filter for IPv4 frames and set the action to "drop."
3. Ensure the first filter (for Destination IP x.x.x.x) is assigned a higher priority than the second filter (for IPv4 frames)

Here's how it works:

- When a frame with Destination IP x.x.x.x is encountered, the first filter applies with no action. Since a filter has already been applied, the lower priority filter will not be applied.
- When an IPv4 frame not matching Destination IP x.x.x.x is encountered, it does not match the first filter, so the second filter applies, and the frame is dropped.

The field *Mask* in *Ingress filter match* specifies which bits in the *Value* field should matter. Users can input “1”s and “0”s in this field where 1 means the corresponding bit in the *Value* field stays the same and 0 means the corresponding bit in the *Value* field could be a “don’t care” value.

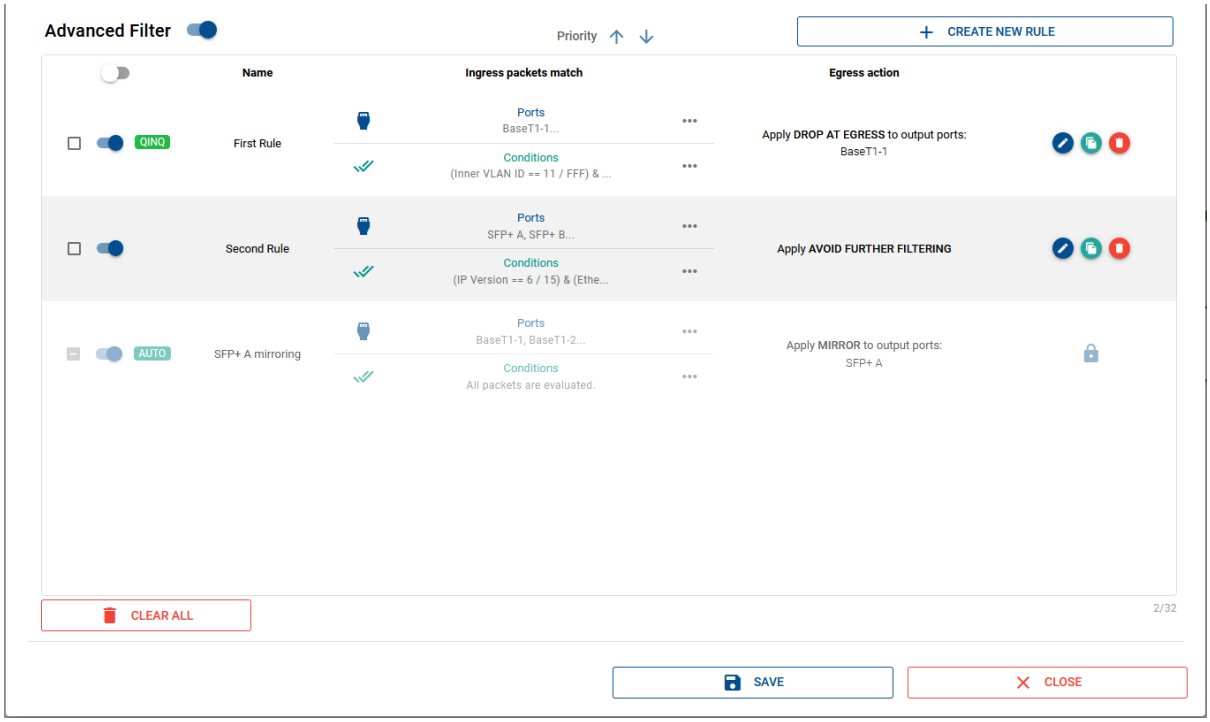


Figure 4-24: Advanced Filter

4.1.8.5 Traffic Shaping

This is a licensed feature. This sub-tab enables the user to fully configure the traffic shaping options that the EES provides.

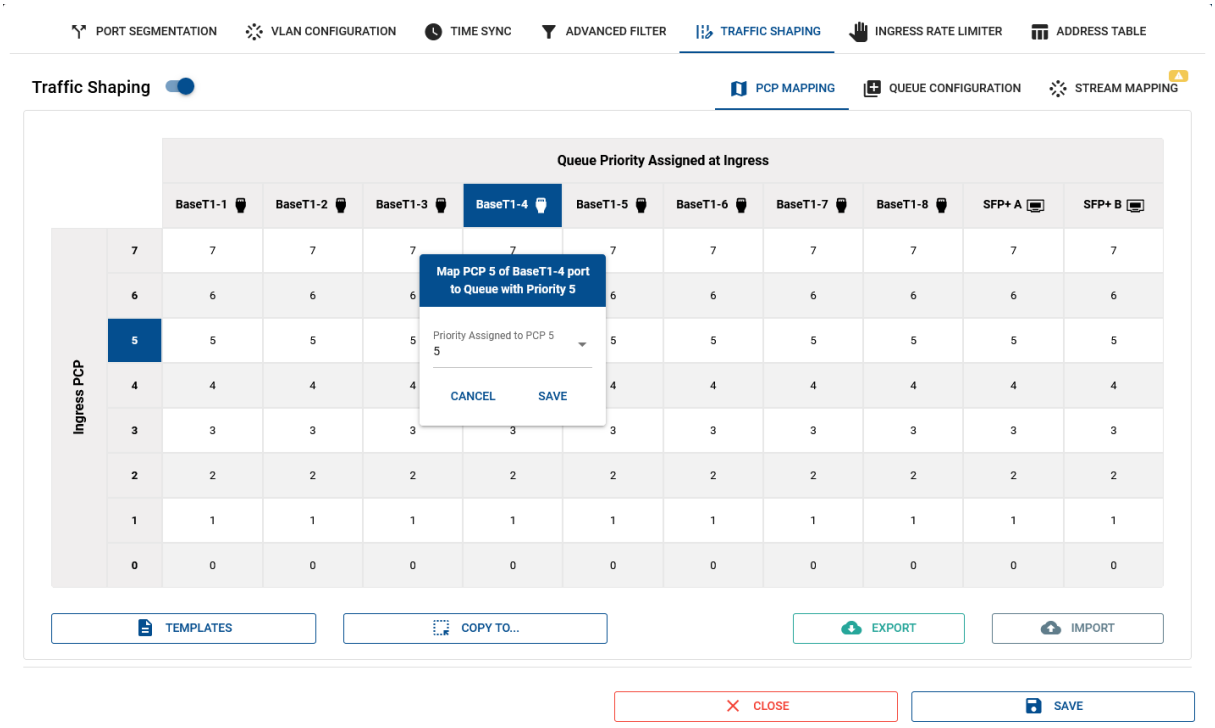


Figure 4-25: Traffic Shaping. PCP Mapping View

The first tab of this function, "PCP MAPPING", configures the carrier priority mapping of the PCP field in the VLAN header. On entry, when a packet arrives at the switch, each packet is assigned a priority from 0 to 7, where 7 corresponds to the highest priority. The switch uses the PCP to assign a different priority depending on the value of the PCP field. The default configuration provides a one-to-one mapping of PCP to priority, but a different mapping can be configured. Clicking on a cell brings up a dialog box that allows us to assign a different queue priority on the corresponding port and PCP value.

The template button offers a couple of pre-configured mapping:

- *The Default template that maps one-to-one the PCP to priority (the default configuration)*
- *An AVB template that sets a predefined mapping typically used in AVB scenarios.*

It is also possible to copy the assignments of a selected port to other ports using the Copy to .... option. Exporting and importing of mappings is possible using the corresponding buttons on the configuration window.

Untagged frames will be assigned to the priority 0.

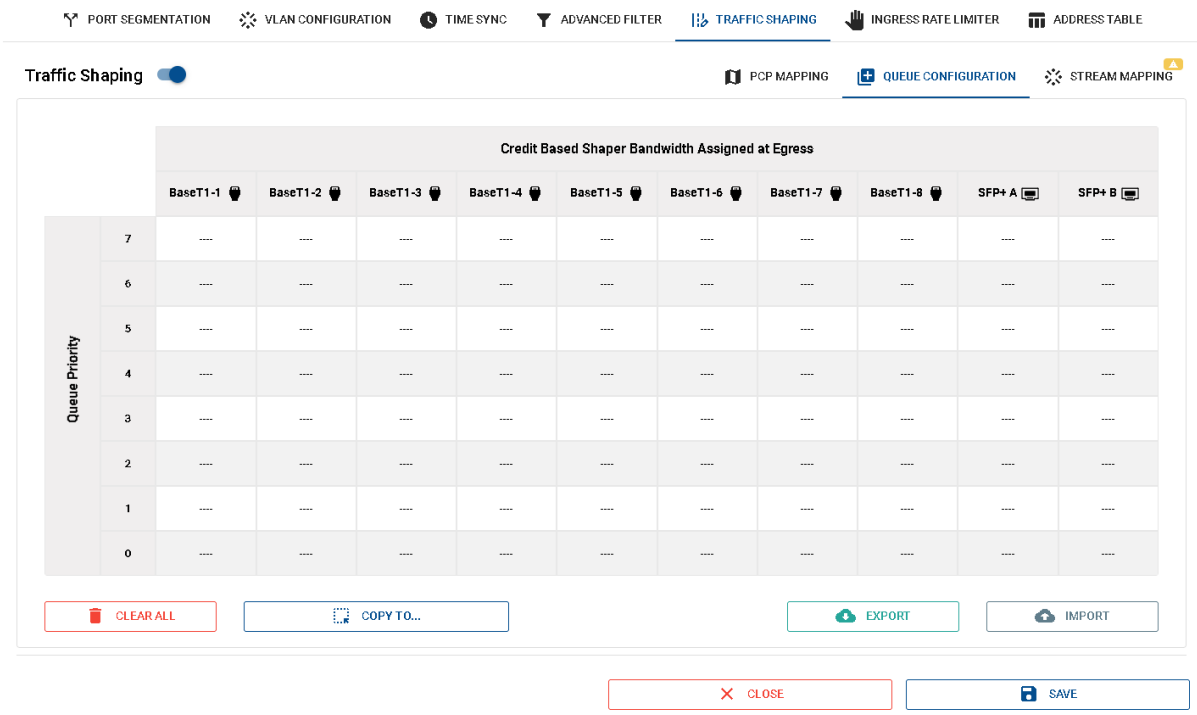


Figure 4-26: Traffic Shaping. Queue Configuration view

Based on the priority assigned at ingress according to the configuration done in the *PCP Mapping* window, the packet is pushed to the corresponding queue of the Egress port.
Each queue output has its own Credit Based Shaper (CBS) that can be configured by clicking onto the field. Here the *CBS* can be enabled, and a *CBS Bandwidth* can be configured.
The scheduler reads the packets from the queues following a "strict" policy, no packets will be sent from a queue if there are packets pending to be sent in a queue of higher priority.

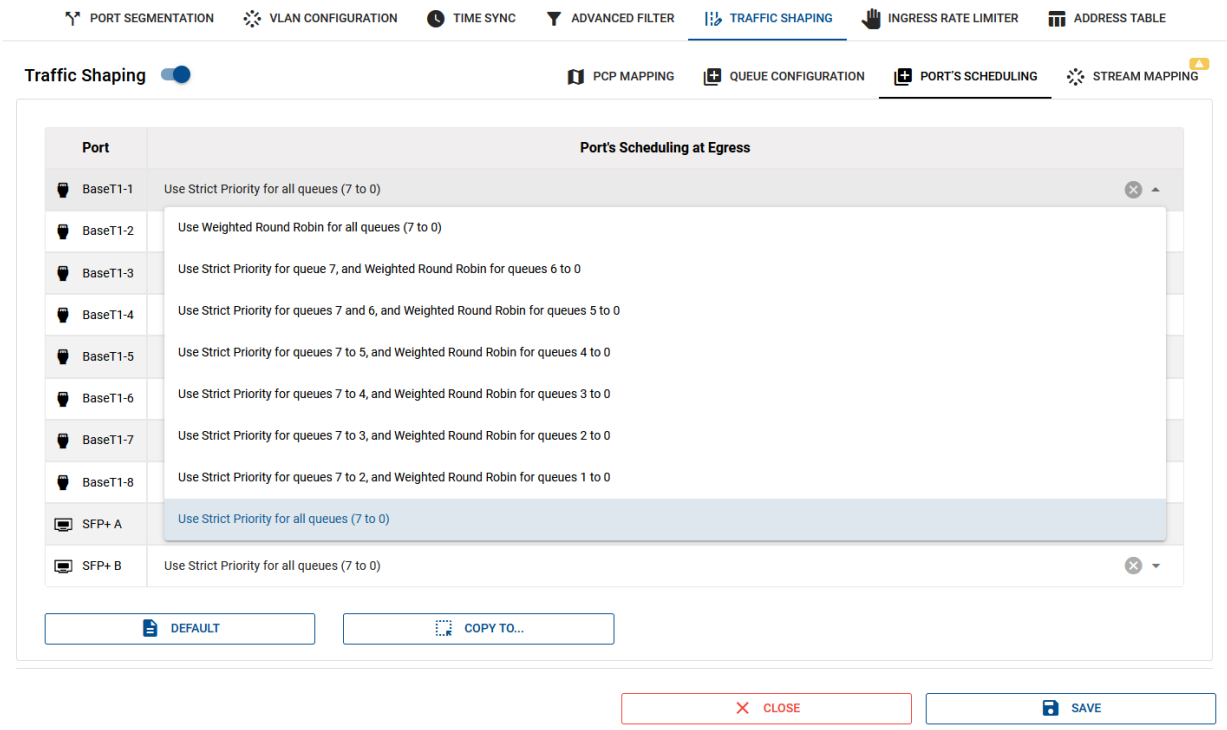


Figure 4-27: Traffic Shaping: Port's Scheduling (Strict Priority and/or WRR)

Weighted Round Robin (WRR) is a scheduling algorithm used in network traffic shaping to manage bandwidth allocation across multiple traffic queues. Unlike simple Round Robin, WRR assigns a weight to each queue, allowing differentiated service levels based on priority or bandwidth requirements.

In WRR-based traffic shaping, each traffic class is assigned to a dedicated queue. Each queue is configured with a specific weight, which determines how many packets it can transmit during each scheduling cycle. This ensures fair bandwidth distribution while allowing differentiated service levels.

Every queue is given an opportunity to transmit packets, with higher-weighted queues receiving more transmission chances. There are a total of 8 queues, numbered from 0 to 7, where Queue 7 holds the highest priority.

WRR operates on a predefined transmission sequence that reflects the relative priority of each queue. The number of transmission opportunities per cycle for each queue is as follows: 7-33, 6-25, 5-17,4-12, 3-6, 2-3, 1-2, 0-1.
This sequence ensures that Queue 7 has the highest number of chances to transmit frames or packets, while Queue 0 has the least.

When a queue is configured with Strict Priority, it will always be serviced before any WRR-based queues. For example, if Queue 7 is set to Strict Priority, it will transmit its packets first, and only after it is empty will the WRR mechanism begin servicing queues 6 to 0.
This feature is highly configurable, Strict Priority can be assigned to one or more queues, ensuring they are serviced first, while the remaining queues will operate under the WRR algorithm. This flexibility allows for optimized traffic handling based on application requirements.

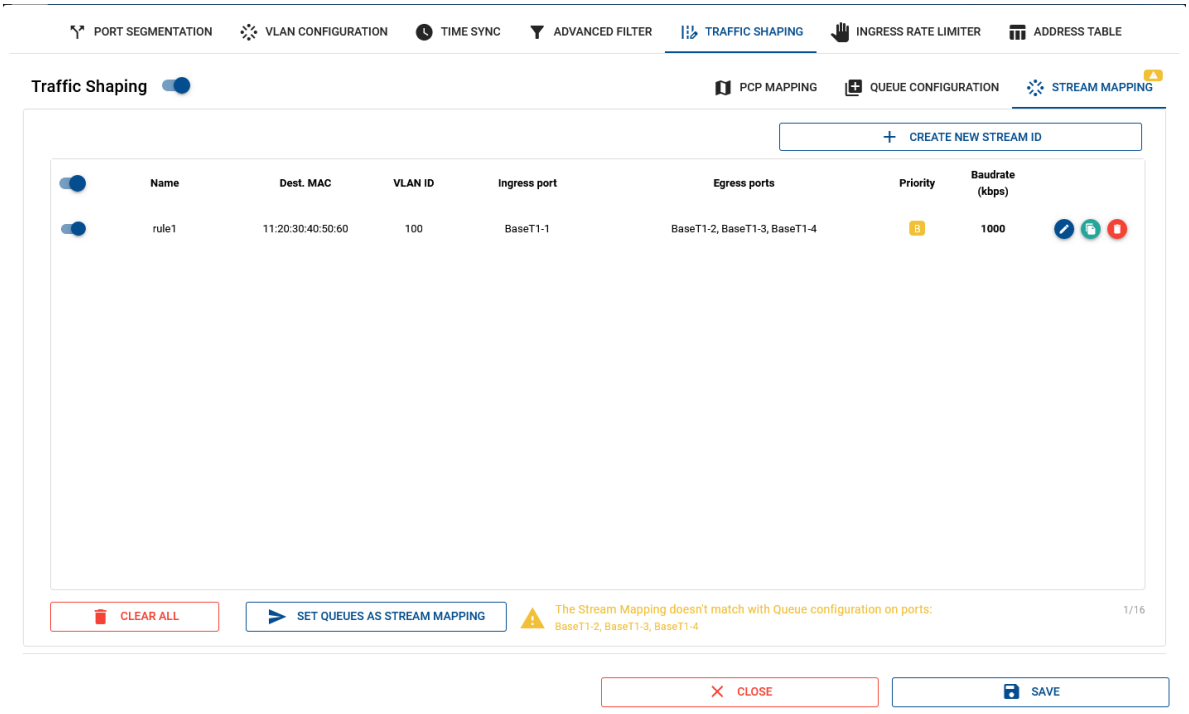


Figure 4-28: Traffic Shaping. Stream Mapping view

For this feature the VLAN Mode needs to be enabled. A notification will inform the user about this necessity.
The Stream mapping view allows to define several streams within the switch. A stream is identified by its ingress port, its destination MAC address and its VLAN ID. For each defined flow, it is necessary to configure the ports over which the packets will be forwarded, the class A or B where A is a higher priority

(a reference to the standard 802.1qat) than B and the maximum bandwidth assigned to. The user can add their own desired values of bandwidth by clicking on , where the user can input Max Payload Size in Bytes, Frame overhead, and Max Frames per Interval. For 1000BASE-T1 ports, the maximum limit is 1000Mbps and for SFP+ ports the maximum limit is 10Gbps, respectively.

The user can define multiple stream reservation rules via *+ Create New Stream ID*.

Once the flows have been defined, it is necessary to translate their configuration into an appropriate Queue Configuration. If the current Credit Base Shaper configuration is not able to match the stream configuration, a warning will be displayed to the right of the *SET QUEUES AS STREAM MAPPING* button. The *SET QUEUES AS STREAM MAPPING* button allows transferring the stream mapping configuration to the queue configuration in two different modes:

- Replace the Queue Configuration according to Stream Mapping: In this mode the CBS of the queues will be replaces to match exactly the configuration on the Stream Mapping.
- Adjust the Queue Configuration to accommodate the Stream Mapping: In this mode the CBS of the queues will be only modified if the bandwidth is not enough to fit all the streams configured.

4.1.8.6 TimeSync

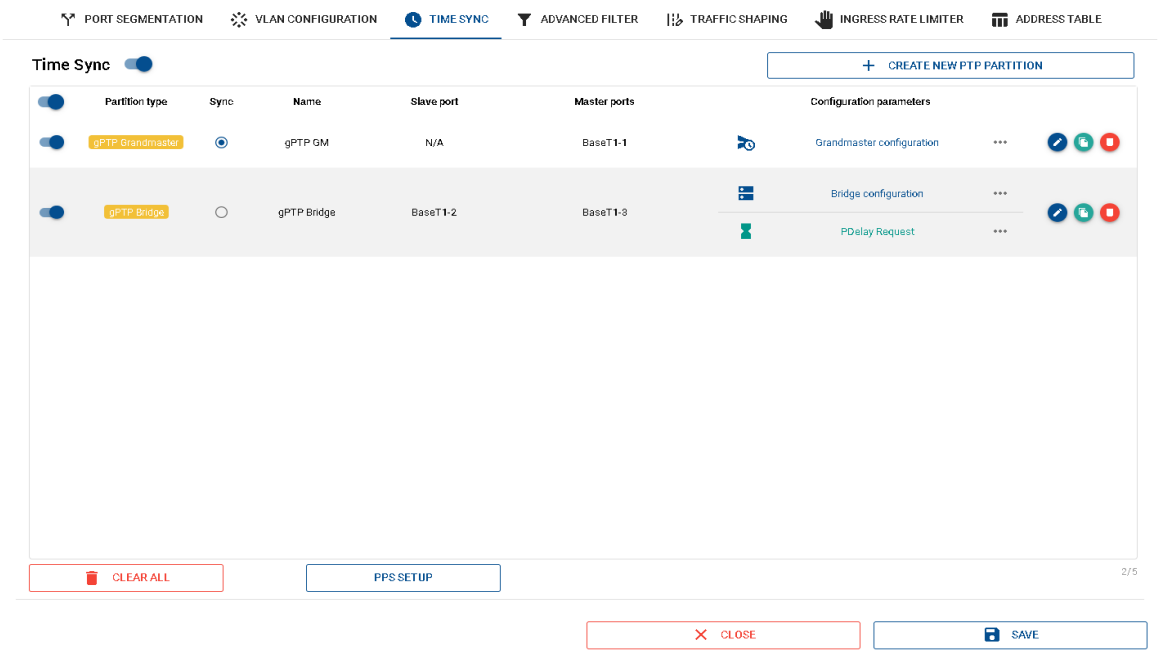


Figure 4-29: TimeSync configuration

To adopt an open approach for the configuration of time synchronization Technica Engineering developed its PTP stack. The implemented stack follows the gPTP standard from AVNU Alliance and partially supports 802.1AS-2020 and IEEE 1588-2008. Using this stack, the user can define more than one bridge slave port per switch. This is achieved in the switch using partitioning inside the switch. The partitioning enables the user to divide the ports of a switch into multiple sections. Each section is an independent partition. The synchronization information received in one partition stays in the same partition. Each partition contains a set of ports. These ports can be Grand Master, Bridge Master, or Bridge Slave. A port can only be assigned to one partition for the same domain number at a time. Up to 5 partitions can be set in the switch.
For Grandmaster partitions (gPTP or PTPv2) the switch provides the option to generate a *Pulse Per Second (PPS)* output signal. The PPS output settings can be configured via the *PPS Setup* menu. The *Sync* column in Figure 4-29 is used to select which Grandmaster partition is used as the synchronization source for the *PPS output*. Only one Grandmaster partition can be assigned to the PPS output at a time. The PPS signal is therefore generated based on the time synchronization information of the selected

partition. After enabling the *PPS output,* the user can configure the pulse width of the generated signal. The available pulse width settings are *1*, *10* or *100* milliseconds.

Concept

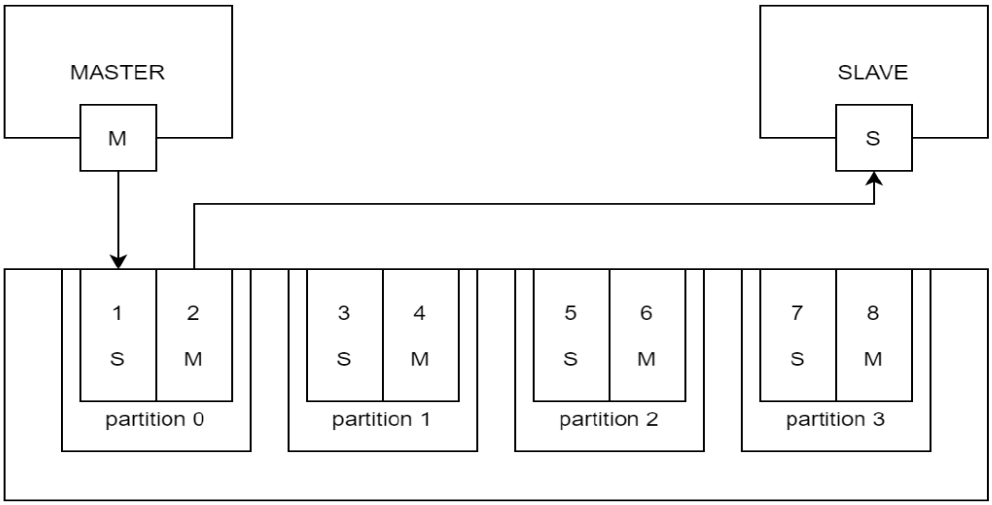


Figure 4-30: Time Sync Partitioning in EES

In the sub-tab Time Sync, the user can configure Time Sync partition(s).

Five types of partitions can be defined in the switch.

- *gPTP Bridge Partition*: This forwards the synchronization information for the slave port to multiple master ports. Only one slave port per bridge can be configured while multiple master ports can be configured.
- *gPTP Grand Master Partition:* The synchronization is generated using the internal oscillator of the switch. The synchronization information can be distributed to multiple master ports.
- *PTPv2 Grand Master Partition:* This is a licensed feature. The synchronization is generated using the internal oscillator of the switch. The synchronization information can be distributed to multiple master ports.
- *PTPv2 Transparent Clock***:** This is a licensed feature. The PTPv2 Transparent Clock forwards synchronization information from the PTPv2 master node to several PTPv2 slave nodes.
- *PTPv2 to gPTP Bridge:* This is a licensed feature. The PTPv2 to gPTP bridge forwards synchronization information from the PTPv2 master node to several gPTP slave nodes
  - There is no support for BMCA (Best Master Clock Algorithm), neither in PTPv2 nor in gPTP.
  - The PTPv2 port receives synchronization, so it can only take the role of slave. The received information is forwarded to all gPTP ports defined in the partition, so these ports can only take the role of master.
  - Only one PTPv2 port per partition. Up to 5 different partitions can be defined in the switch.

These are the available fields the user can use to configure his partition. Not all fields described in the following are available for every partition type:

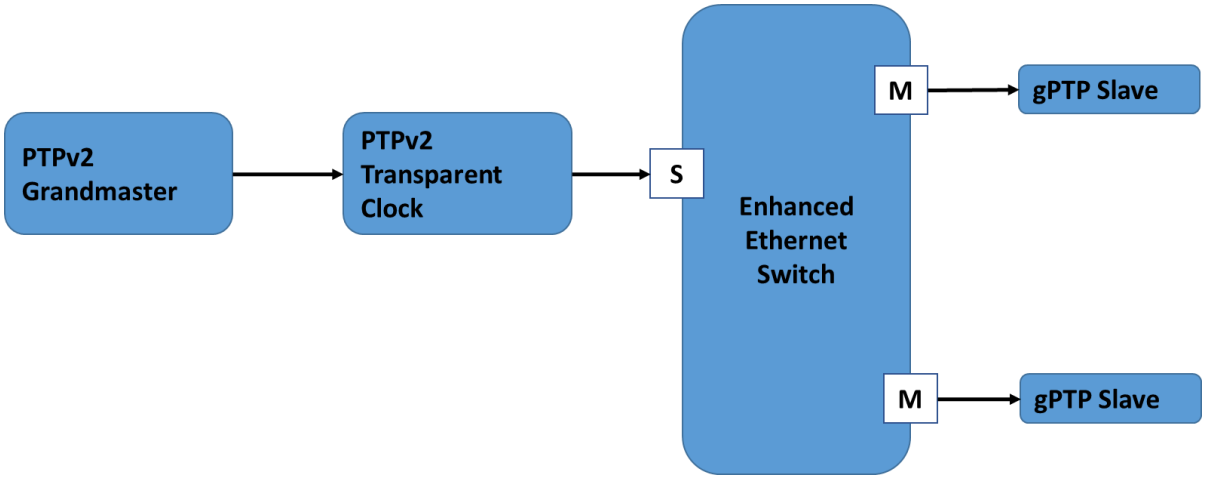


Figure 4-31: PTPv2 to gPTP Bridge

These are the available fields the user can use to configure his partition. Not all fields described in the following are available for every partition type:

Set partition name: The user can name the partitions created in the switch.

Select partition type*:* Here the user can create either a Bridge partition or a Grand Master partition.

Select transport layer*:* The user can choose the physical transporter layer for the partition. The transport layer can be selected between Ethernet (Layer 2) and IP4/UDP (Layer 3). (Only available for PTPv2 partitions)

Destination Mac address used in PTPv2 Layer 2: From release of R.017.000 onwards the MAC destination addresses for PTP messages are implemented as follows:

- gPTP: Implementation according to standard.
  - For all PTP messages: 01:80:C2:00:00:0E.
- PTPv2 Layer 3: Implementation according to standard.
  - For all PTP messages (except Peer Delay): 01:00:5E:00:01:81.
  - For Peer Delay Messages: 01:00:5E:00:00:6B.
- PTPv2 Layer 2: Implementation according to standard we have 3 options to increase interoperability:
  - Option 1: For all PTP messages: 01:80:C2:00:00:0E.
  - Option 2: For all PTP messages: 01:1B:19:00:00:00.
  - Option 3: For all PTP messages (except Peer Delay): 01:1B:19:00:00:00 / For Peer Delay Messages: 01:80:C2:00:00:0E.

Select Delay Mechanism: The user can choose the delay mechanism between *End To End* and *Peer To Peer*

Select slave port: Here the user can select slave port of the time sync partition

Domain number for the slave port*:* Configure the domain number used for the PTP communication in this partition on the slave port.

Use the same Domain Number on the slave and master ports: When enabled, the domain number configured on the slave or master port(s) is used in both.

Select the slave port: Select the port that needs to be connected to the time sync master for the switch to work as a bridge.

Enable VLAN on the slave port: Once enabled, the user can define the *VLAN ID* and VLAN egress priority used for egressing PTP frames on the slave. On ingress only PTP frames matching the VLAN ID will be utilized by the slave port.

Forward Announce messages to master ports: When enabled, the announce message received on the slave port will be forwarded to the master port(s) and sent out.

Mirroring PTP messages egressing from the slave port: This enables the mirroring of egressing PTP frames from the slave port to a selected port that is not used by any partition.

Select master ports: Here the user can select one or multiple ports working as PTP master port(s) for the time sync partition.

Domain number for the master port: Configure the domain number used for the PTP communication in this partition on the master port.

Enable VLAN on the master port: Once enabled, the user can define the VLAN ID and *VLAN egress priority* used for egressing PTP frames on the master port(s). On ingress only PTP frames matching the VLAN ID will be utilized by the master port(s).

Mirroring PTP messages egressing from the master port: This enables the mirroring of egressing PTP frames from the master port(s) to a selected port that is not used by any partition.

Transmit PDelayReq messages on master ports: Enabling this option forces the master port to transmit PdelayReq messages and compute the mean path delay of the link. The measured mean path delay can be viewed in the Dashboard.

Number of Sync messages lost to trigger a timeout: Here the user can define the number of sync messages between 1-5 that must be lost to consider a timeout event.

Interval period for gPTP-capable messages sent on all ports: Here the user can enable the sending of the gPTP-capable message. Currently, the following selections can be made.

- 500 milliseconds
- 1 second
- 2 seconds
- 4 seconds
- 8 seconds

Number of gPTP-capable messages lost to trigger a timeout: Here the user can define the number of gPTP-capable messages between 1-5 that must be lost to consider a timeout event.

Proxy operation (only in Bridge mode): If the synchronization is lost in the slave port, the bridge must enter "Proxy Mode". In this mode, the master ports must keep sending synchronization messages using the latest valid information and updating it with its internal clock. When "Proxy Mode" is enabled, the user can define the following parameters.

- If the *Proxy operation* is set to true, the proxy operation will be triggered if the synchronization on the slave is lost
- If the *Proxy operation* is set to false, the partition will never jump to the proxy operation

Default Sync interval on master ports: If proxy mode is enabled, then the default sync interval shows the synchronization message interval used on master ports until a valid synchronization is received on the slave port. The master port sends a message to synchronize. The interval for the synchronization message can be set here. The current selection offers the following choices.

- 31.25 milliseconds

- 62.5 milliseconds
- 125 milliseconds
- 250 milliseconds
- 500 milliseconds
- 1 second

Default Date / Time: The date/time sent by master ports when working in proxy mode and no valid synchronization information has been received on the slave port.

Users can configure other parameters for fine-tuning, but they have default values in case they are not changed. These parameters are:

- *Startup timeout (in seconds):* This is the wait time to jump to proxy mode if at startup the slave port does not receive any synchronization information

Initial Delay Request interval (only in Bridge Mode): The slave port sends a message to calculate path delay. The interval for a path delay request message can be set here. If 1 second is selected it means the slave port sends a path delay request every second until the system considers the measurements to be stable. Currently, the following selections can be made.

- 500 milliseconds
- 1 second
- 2 seconds
- 4 seconds
- 8 seconds

Operational Delay request interval: After the system considers the measure of path delay request to be stable, the user can define an interval for transmitting path delay requests in this field. An operation path delay request enables the user to define a time for transmitting path delay requests after the system has achieved stability thus avoiding overloading the system with unnecessary path delay requests. The current selection offers four choices

- 500 milliseconds
- 1 second
- 2 seconds
- 4 seconds
- 8 seconds

Default mean path delay (in nanoseconds): When setting the Initial PDelay request interval to *Do not send PDelayReq messages*, the user has the option to set a Default mean path delay (in nanoseconds)

Number of PDelayReq with no response to trigger a timeout: Here the user can define the number of PDelayReq messages between 1-5 that must be unanswered to consider a timeout event.

Mean path delay mechanism: Here the user can choose the mechanism that will be used to measure the mean path delay. The user can choose between Use Peer to Peer with domain 0 (802.1AS-2011 / 802.1AS-2020) and *Use CMLDS (802.1AS-2020)*. When multiple partitions share the same slave port with different domains, CMLDS is recommended.

Initial Date / Time: The date/time sent by master ports when it starts operating as grandmaster can be defined here.

Initial Time, nanoseconds part: The nanoseconds parts of the Initial Date/Time.

Interval period of Sync messages sent on master ports: This defines the synchronization interval: The master port sends a message for synchronization. The interval for the synchronization message can be set here. The current selection offers the following choices.

- 31.25 milliseconds
- 62.5 milliseconds
- 125 milliseconds
- 250 milliseconds
- 500 milliseconds
- 1 second

Announce interval period on master ports: This defines the announce message interval: The master port sends a message announce message. The interval for the announce message can be set here. The current selection offers the following choices.

- Disable announce message (default)
- 500 milliseconds
- 1 second
- 2 seconds
- 4 seconds
- 8 seconds

UTC offset valid: If enabled the flag "currentUtcOffsetValid" is set in the announce message.

UTC offset in seconds: The value set in the "currentUtcOffset" field of the announce message.

Clock Priority 1, Clock Priority 2, Clock Class, Clock Accuracy, Clock Variance, and Time Source: The values set in the "grandmasterPriority1", "grandmasterPriority2", "grandmasterClockQuality" and "timeSource" fields of the announce message.

4.1.8.7 Ingress rate limiter

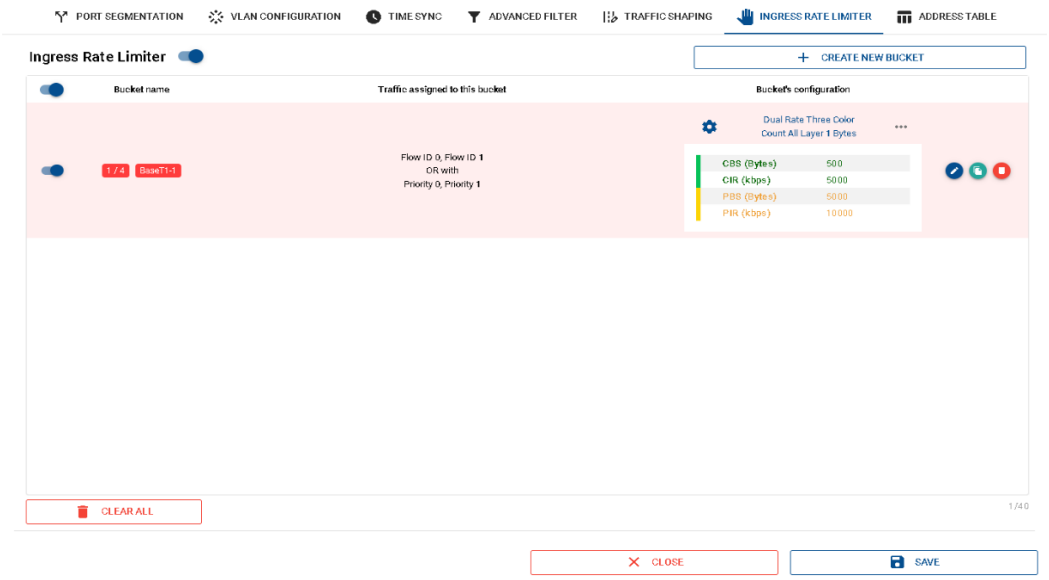


Figure 4-32: Ingress rate limiter

This is a licensed feature. For the switch receiving multiple streams simultaneously, per-stream filtering and policing enable a bridge or endpoint component to detect whether components in the network are conforming to the agreed rules. For example, a node gets allocated a certain amount of bandwidth and

when this bandwidth is exceeded due to a component failure, or malicious act, appropriate action can be taken to protect the network. This feature includes procedures to perform frame counting, filtering, policing, etc. Policing and filtering functions are useful for detecting and enabling the elimination of disruptive transmissions, thus improving the robustness of the network.

In the Enhanced Ethernet switch -up to 4 available FlowIDs per port can be defined so up to 4 buckets per port are available. Frames are selected using an advanced filter rule and marked with a FlowID which is used as an ingress filter rule by ingress rate limiter.

Concept

When a stream ingress the switch, it passes through multiple blocks to be serviced by an ingress rate limiter. These blocks are described below

1. Stream marking: The ingress stream enters the stream marking block. Here the TCAM Flow ID rule defined in the advanced filter isolates the stream based on the parameters defined. The parameters that are used to isolate the traffic are ingress port, source MAC address, destination MAC address, VLAN, priority, egress action, and TCAM Flow ID. In automotive applications streams are defined statically in each one of the resources in the network. The advanced filter feature, implemented in the switch, is used to mark a specific stream to be metered by an ingress rate limiter flow meter.
2. Flow metering: The stream from the stream marking block enters the flow metering block. Here users can define token buckets to limit the ingress rate. These token buckets can be defined using the following parameters.
   - Committed burst size (CBS)
   - Committed information rate (CIR)
   - Excess burst size (EBS)
   - Excess information rate (EIR).

The ingress rate limiter limits the bandwidth through token buckets. Three different types of token buckets can be configured.

1. Two-color single rate
2. Three color single rate
3. Three color double rate

Two-color single rate

The user can define the size of the bucket in Bytes in the field committed burst size(CBS). All the ingress traffic can be accumulated in the bucket and egressed out by the rate defined through the committed information rate(CIR). If the ingress traffic is more than the size of the committed bucket rate, the bucket is over-filled and excessive traffic is dropped.

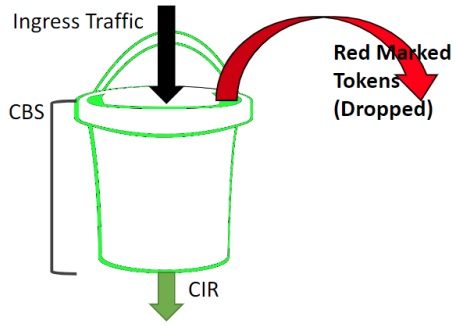
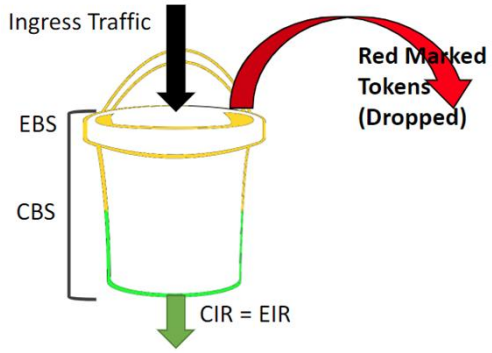


Figure 4-33: Two-color marker

Three color single rate

The user can define the size of the bucket in Bytes in the field committed burst size(CBS). The user can also define excess burst size (EBS). This means in case of congestion the yellow packets (EBS) will not be serviced. If there is no congestion, all the ingress traffic (CBS and EBS) will be serviced. In case of no congestion, all the ingress traffic can be accumulated in the bucket and egressed out by the rate defined through committed information rate (CIR). Since the bucket has only one egress, the committed information rate (CBS) should be equal to the excessive information rate (EBS).



Three color double rate

Figure 4-34: Three color single rate

The user can define the size of the bucket in Bytes in the field committed burst size (CBS). The user can also define excess burst size (EBS). This means in case of congestion the yellow packets (EBS) will not be serviced. If there is no congestion, all the ingress traffic (CBS and EBS) will be serviced. In case of no

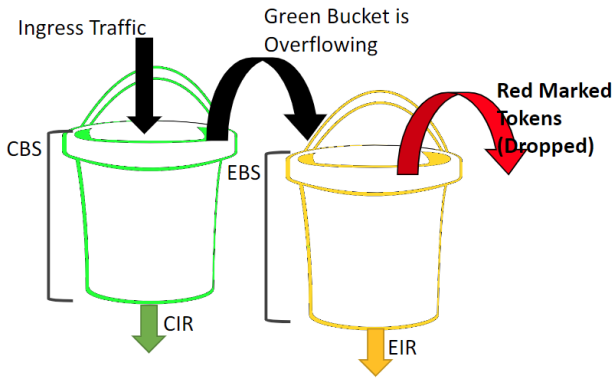


Figure 4-35: Three color double rate

congestion, all the ingress traffic can be accumulated in the corresponding buckets and egressed out by the rate defined through the committed information rate (CIR) and the excessive information rate (EIR).

Configuration

In the sub-tab *Ingress rate limiter*, the user can configure the following parameters

Select ingress port: The user can select the ingress port where the limiting of the rate is desired.

Select ingress traffic: The flow meters are implemented as token buckets. The user can select the FlowIDs to be applied to the bucket. More than one stream/priority can be metered by a single bucket.

Priority assignment mode: *The user can select between:*

- The traffic type selected above or the packets with the priorities selected below are assigned to this bucket.
- The traffic types above selected, which also have the priorities selected below, are assigned to this bucket.

Select priorities: The user can select priorities between 0-7 to be applied to the buckets.

Set bucket parameters: The following parameters can be defined to form a bucket

- *Committed burst size (CBS):* This defines the size of the bucket in Bytes
- *Committed information rate (CIR):* This defines the number of tokens periodically egressing the bucket in kbps.
- *Peak burst size (PBS):* This defines the sum of committed burst size and excess burst size in Bytes.
- *Peak information rate (PIR):* This defines the sum of excess information rate (EIR) tokens periodically egressing the bucket and the committed information rate (CIR).

4.2 MACsec Configuration tab

On the tab “MACsec Configuration “, the user can configure various MACsec and MKA parameters.

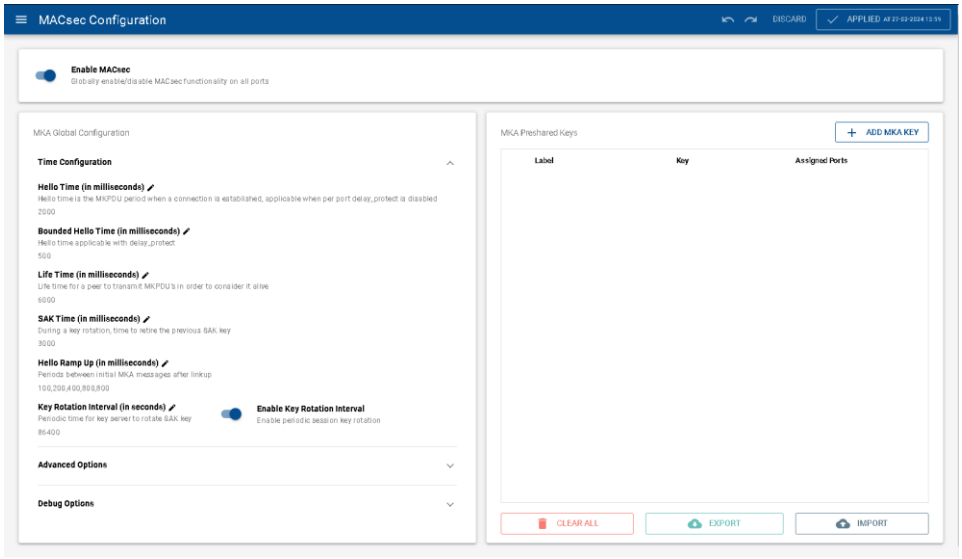


Figure 4-36: MACsec configuration parameters



This option enables the user to enable or disable MACsec functionality on all ports. By default, the value is disabled, thus enabling the user to configure MACsec parameters on this tab as well as on the BASE-T1 ports.

4.2.1 MKA Global configuration

The user can configure multiple parameters regarding MKA. The following options can be configured.

4.2.1.1 Hello Time (in milliseconds)

Hello time is the MKPDU period (in milliseconds) when a connection is established. It is applicable when *mka.delay_protect* is disabled in the BASE-T1 ports. By default, the value is set to 2000 milliseconds. The possible range of values for this field is between 0 to 100000 milliseconds.

4.2.1.2 Bounded Hello Time (in milliseconds)

Hello time applicable with delay_protect (in milliseconds). By default, the value is set to 500 milliseconds. The possible range of values for this field is between 0 to 100000 milliseconds.

4.2.1.3 Live Time (in milliseconds)

Lifetime (in milliseconds) for a peer to transmit MKPDUs to consider it alive. By default, the value is set to 6000 milliseconds. The possible range of values for this field is between 0 to 100000 milliseconds.

4.2.1.4 SAK Time (in milliseconds)

This field mentions during a key rotation, the time to retire the previous SAK key (in milliseconds). By default, the value is set to 3000 milliseconds. The possible range of values for this field is between 0 to 100000 milliseconds.

4.2.1.5 Hello Ramp Up (in milliseconds)

This option mentions the periods between initial MKA messages after linkup (in milliseconds). Up to 8 values can be configured in this field. By default, the value is set to [100, 200, 400, 800, 800]. The possible range of values for this field is between 0 to 100000 milliseconds.

4.2.1.6 Key Rotation Interval (in seconds)

This option mentions the time for the key server to rotate the SAK key. By default, this interval is set to 86400 seconds, and the key rotation is enabled. The user has the possibility to disable the key rotation interval.

4.2.1.7 Unique Port ID in the Secure Channel Identifier (SCI)

Enabling this option enables the port ID in the Secure Channel identifier to be unique and assigned consecutively for each port

4.2.1.8 Transmit NULL XPN

This option determines whether to transmit a null XPN parameter when working without extended packet numbers. By default, the option is enabled.



IEEE802.1X-2020 Figure 4-10 note b states this parameter's PN is transmitted as zero when the cipher suite does not support extended packet number, which corresponds to setting "on".

4.2.1.9 Disable non successful participants

This option determines whether to stop non successful participants once principal elected (IEEE 802.1 X deviation). By default, this feature is enabled.

4.2.1.10 Disable EAPOL/MKA packets forwarding

This option determines whether disable forwarding for IEEE 802.X reserved MAC 01:80:C2:00:00:03. By default, this feature is disabled.

4.2.1.11 Fail Safe Mode

Enabling this feature gives the user the option to select *ports to be monitored* for a configured *MKA Negotiation Timeout (in milliseconds)*. Upon fulfilling a trigger condition, the switch will check if the MKA negotiation was successfully completed within the defined timeout period for the selected ports. If not, MACsec will be disabled for all ports and a warm reset will be performed.

4.2.1.12 Log MKA messages to Telnet server

When this option is enabled the logging information provided by the MKA stack can be read if the user uses a telnet connection to the IP address of the device. (This option is **only** intended for developers and MACsec experts.)

4.2.1.13 Mirror MKA Packets

When this option is enabled, the user can *select ports to be mirrored* and a *mirroring port*. MKA packets will then be mirrored to this mirroring port.

4.2.2 MKA Preshared Keys

This table can store all the keys that can be assigned to MACsec ports. Up to 32 keys can be stored in this table. The user can import and export the MKA keys using this menu as well. Each Key Item stores the following fields:

- **label**: a string of up to 16 characters in length assigned by the user
- **cak**: a vector that stores 16 or 32 bytes. Any size other than these 2 is considered an error.

- ckn: a vector that stores up to 32 bytes and it cannot be empty.

4.3 Control panel tab.

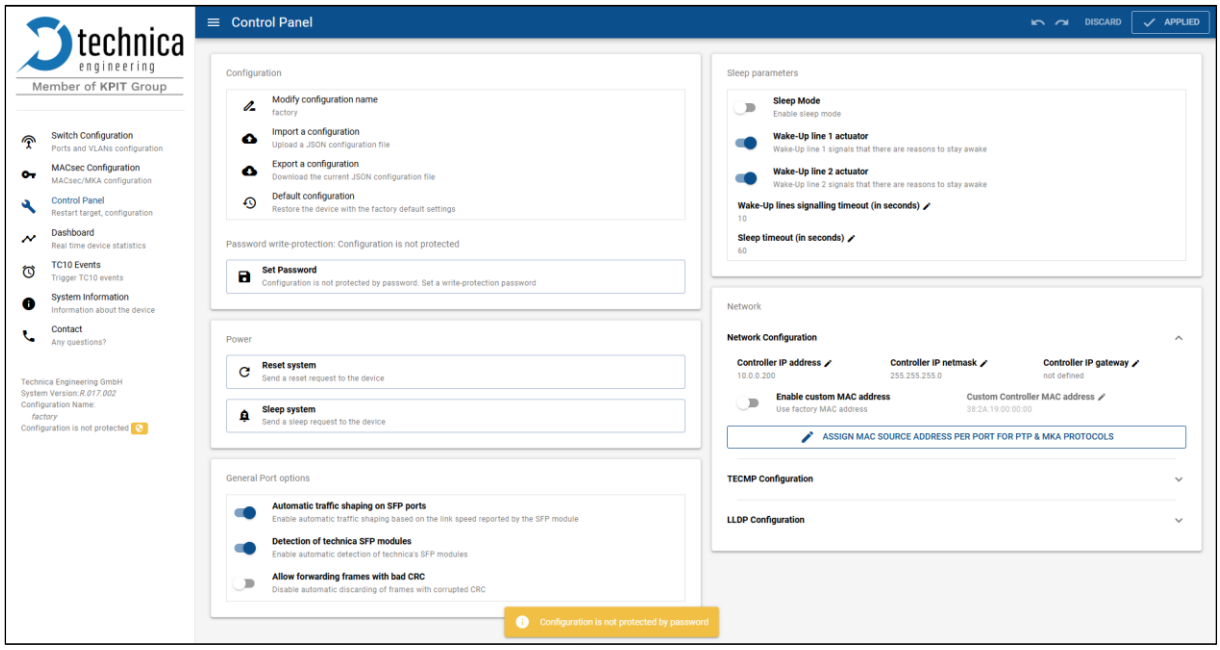


Figure 4-37: Control Panel Tab

4.3.1 Configuration

Users can make general settings of the configuration in the switch in this tab.

4.3.1.1 Modify configuration name

You can rename the configuration to any name of your choice. After saving, the new name is applied.

4.3.1.2 Import a configuration

Through this option, the user can import a configuration from the computer to the switch in .json format.

4.3.1.3 Export a configuration

Applied changes can be exported to the computer through this option. The exported configuration is in *.json file extension format.

4.3.1.4 Default configuration

Through this option, the switch is reset to the default configuration. The IP address remains the same and is not reset to default.

4.3.2 Password write-protection.

The user can protect the configuration with a password (min. 8 digits), so only authorized persons can modify the configuration. The configuration can still be exported but for uploading another configuration the password has to be inserted. The password can be changed or cleared on the webpage by inserting the current password. With the factory HW button on the EES the configuration can be set to default what also removes the password protection.

4.3.3 Power

This section offers settings for power and sleep functions.

4.3.3.1 Reset System

When this option is selected, the device can trigger a power reset.

4.3.3.2 Sleep System

This option sends a sleep request to the device.

4.3.4 General port options

This section offers settings for the ports.

4.3.4.1 Automatic traffic shaping on SFP ports

When this feature is enabled the EES automatically sets traffic shaping based on the link speed reported by the SFP module for the SFP port. However, if the user manually sets traffic shaping for the SFP ports and the configured value is lower than the reported speed by the SFP module the manually set value will be used.

4.3.4.2 Detection of Technica SFP modules

When Technica SFP module detection is enabled, the EES firmware automatically configures the appropriate SFP mode when a Technica SFP module is inserted. The user interface also provides simplified controls to easily modify SFP parameters such as speed and master/slave mode.

4.3.4.3 Allow forwarding frames with bad CRC.

By default, the switch discards any frames with CRC errors upon ingress. This is the safest operating mode, ensuring that corrupted frames are not propagated through the network. When the "Allow forwarding frames with bad CRC" option is enabled, frames with invalid CRC values are not discarded and are forwarded like standard frames. This feature is intended solely for testing and diagnostic purposes. Enabling it may compromise data integrity, mask underlying network issues, and introduce instability in production environments.

4.3.5 Sleep parameters

This section offers settings about sleep and wake-up features.

4.3.5.1 Sleep mode

When disabled, the device is working without entering into sleep mode. If this option is enabled, the device will go to sleep if no activity is recognized according to the observed ports. The user must use a Wakeup line to wake the device up if the device goes to sleep.



Please see the configuration of each port in CHAPTER 3.1.6, 3.1.7, 3.1.8, 3.1.9 and 3.1.10.

4.3.5.2 Wake-Up line 1 actuator

There are two WakeUp Lines. Both WakeUp lines can be output.

- If this option is enabled, the wakeup line provides the same voltage as the voltage on the powerline (kind of loop-through). It provides power as long as there is a reason to stay awake
- If this option is disabled, this wakeup line is only input and wakes up the device when supplied

4.3.5.3 Wake-Up line 2 actuator

There are two WakeUp Lines. Both WakeUp lines can be output.

- If this option is enabled, the wakeup line provides the same voltage as the voltage on the powerline (kind of loop-through). It provides power as long as there is a reason to stay awake
- If this option is disabled, this wakeup line is only input and wakes up the device when supplied

4.3.5.4 Wake-Up lines signaling timeout (in seconds)

Here users can define the time in seconds, which must go by before the actuator of the wakeup lines is set to ground. The default value is 10 seconds.

4.3.5.5 Sleep timeout (in seconds)

Here users can define the time in seconds, which must go by with no communication on the supervised ports before the switch goes to sleep. Reasonable values are between 15s and 300s.

4.3.6 Network

This section offers settings about the Network, TECMP and LLDP configuration used for the switch.

4.3.6.1 Controller IP Address

The IP address assigned to the switch can be changed here. By default, the IP address is 10.0.0.200.

4.3.6.2 Controller IP netmask

Defines the network and host portions of an IP address. It helps determine which devices are on the same network. Example: 255.255.255.0 means the first three numbers are the network, and the last is for hosts.

4.3.6.3 Controller IP gateway

The device (usually a router) that connects your local network to other networks, including the internet. It acts as an exit point for data traffic. Example: 192.168.1.1.

4.3.6.4 Enable custom MAC address

Enables the use of a custom MAC address instead of the factory MAC address.

4.3.6.5 Controller MAC address

Here the user can configure the controller MAC address that will be use when enabling custom MAC address.

4.3.6.6 Assign different MAC addresses per port for PTP and MKA protocols

This feature allows users to assign different MAC addresses to all ports of the switch (except the host port). These custom MAC addresses are only used as the source MAC address for MKA (MACsec Key Agreement) and PTP (Precision Time Protocol) messages. The user can decide if he wants to enable the feature for all ports or only for selected ports. By default, every port has the default MAC address of the EES assigned to it, but the MAC address can be configured for every port individually or can be configured ascending for every port by using the button *ASSING CONSECUTIVE MACs FROM BASE-T1-1*.

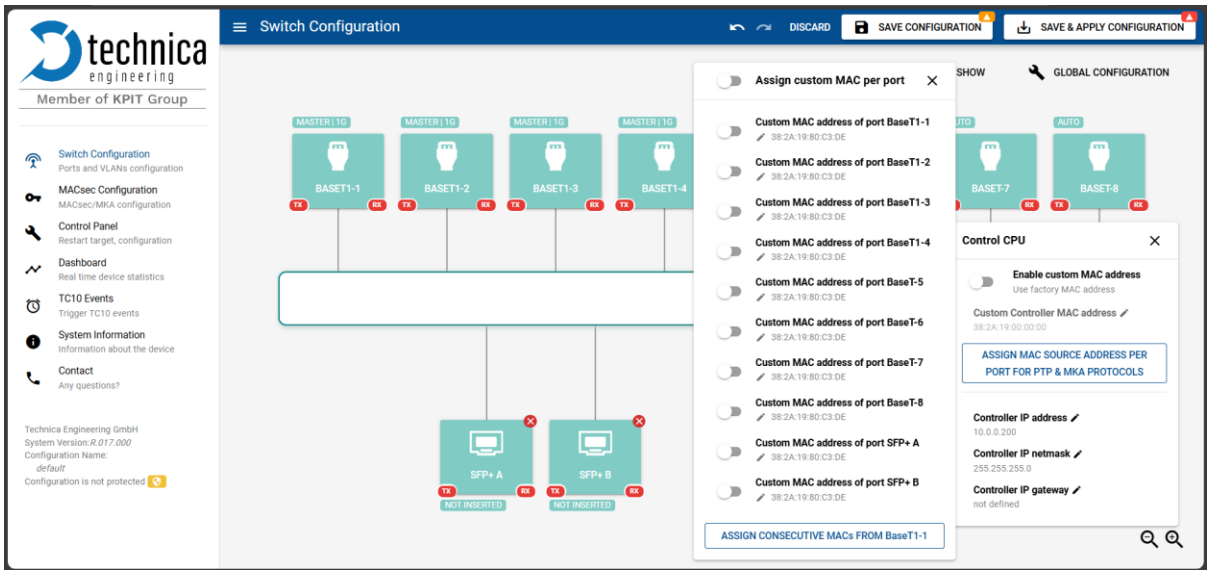


Figure 4-38: Assign different MAC addresses per port for PTP and MKA protocols

With the introduction of this feature in addition to feature *Enable custom MAC address* there are three different configuration levels:

- Use the MAC assigned at production in any message generated in the EES microcontroller.
- Use the Custom MAC configured by the user in any message generated in the EES microcontroller.

- Use the user-configured MAC per port in MKA/PTP messages; for the rest, use the production MAC or the custom MAC depending on the above points.

4.3.6.7 TECMP Unit identifier

The TECMP Unit Identifier is used for TECMP status and configuration messages. By default, the identifier is 0x00BC (EES1). Four possible values are assigned to the EES (from 0x00BC to 0x00BF both included). Besides these values, any other value can be entered, but a warning will be shown indicating that a non-standard value is used.

4.3.6.8 TECMP Ports

This field allows us to configure on which ports TECMP messages will be sent. By default, only the Control port is configured, but any other combination of ports is possible.

4.3.6.9 Enable LLDP

LLDP (Link Layer Discovery Protocol) is a network protocol used for device discovery on a local network. It allows network devices (like switches, routers, and computers) to share information about themselves, such as identity, capabilities, and neighbors. LLDP operates at the data link layer (Layer 2) and is vendor-neutral, making it widely used for network management and troubleshooting.

4.3.6.10 Time Interval (in seconds)

The switch periodically transmits LLDP frames to its neighbors for having the network discovery information up-to-date. The interval between each LLDP frame is determined by the Tx Interval value. Valid values are restricted to 5 - 32768 seconds.

4.3.6.11 TxHold (TTL=121 seconds)

Each LLDP frame contains information about how long the information in the LLDP frame shall be considered valid. The LLDP information valid period is set to Tx Hold multiplied by Tx Interval seconds. Valid values are restricted to 2 - 10 times

4.3.6.12 Active Ports

Actively sends LLDP advertisements (LLDPDUs) to announce its presence and capabilities to neighbors.

4.3.6.13 Passive Ports

Listens for LLDP advertisement from other devices but does not send its own.

4.4 System information tab

On the tab “System Information “, some status information about the device is displayed.

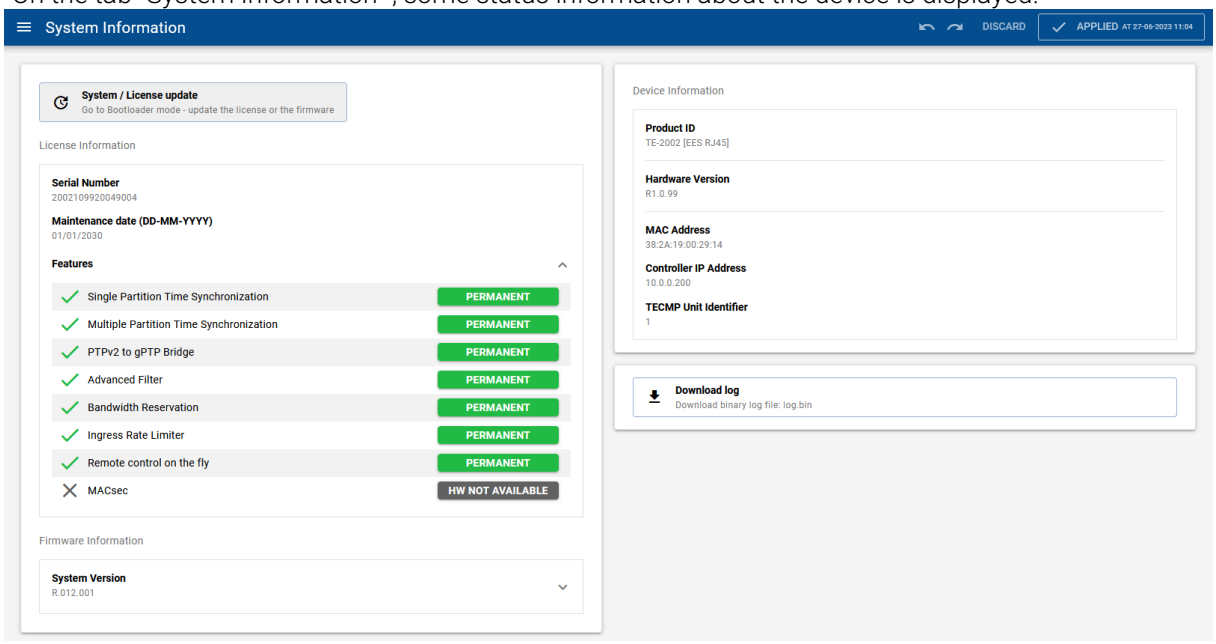


Figure 4-39: System Information Tab

4.4.1 License information

System update: This option is used to update the firmware on the device. The process of updating a firmware/license is explained in CHAPTER 5. When clicking on this option, the device goes into bootloader mode and loses its normal operation. The bootloader mode can also be verified by the fast toggle of the status LED (see CHAPTER 3.1.11) which in normal mode is in slow toggle mode. If this option is selected by mistake, please select “go back to application mode” to exit the bootloader mode.

Serial number: Serial number that identifies the device unequivocally.

Maintenance date (YYYY-MM-DD): The date when the license is valid.

Maintenance date (YYYY-MM-DD): The date when the license is valid.

Features: The list of all the features that are covered under the license is mentioned here. There can be several modes of license.

1. *Unlicensed:* No license is available for this feature.
2. *Temporal:* The license is granted for the limited amount of time mentioned on the webpage.
3. *Permanent:* The license is granted permanently for this device.
4. *Expired:* The license for the feature mentioned is expired and the feature cannot be used anymore.

Currently, Enhanced Ethernet Switch has the following features that need a license.

- Single Partition Time Synchronization
- Multiple Partition Time Synchronization
- PTPv2 to gPTP Bridge
- Advanced Filter
- Bandwidth Reservation
- Ingress Rate Limiter
- Remote control on the fly (See CHAPTER 6)
- MACsec

4.4.2 Device information

Product Id: This shows the name of the device.
Hardware version: The version of your hardware of the switch is displayed here.
MAC address: This MAC address should be the same as on the label at the bottom of the device.
Controller IP address: The user can change the IP address of the switch, in this field. If the user wants to use multiple devices in one network, the user must configure a unique IP address for each device.

If you forgot the IP address of your device, you could reset it to default settings as described in CHAPTER 3.1.2.

TECMP unit identifier: The TECMP Unit Identifier is used for TECMP status and configuration messages. By default, the identifier is 1.

4.4.3 Core Feature Package

When purchasing an EES the core feature package is provided by default. The core feature package includes:

- Single Partition Time Synchronization
- Advanced filter
- Remote control on the fly

4.4.4 Advanced Feature Package

The core feature package can be extended by the advanced feature package that includes:

- Multiple Partition Time Synchronization
- PTPv2
- Bandwidth Reservation
- Ingress Rate Limiter

4.4.5 MACsec Feature Package

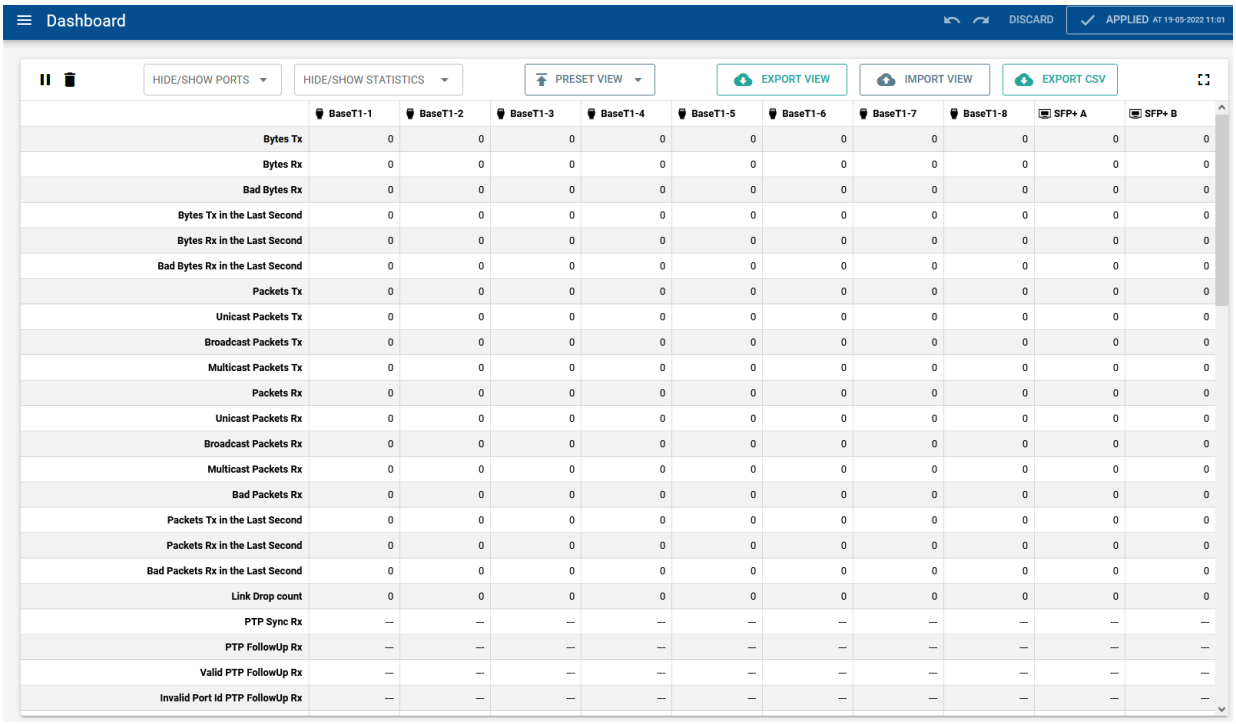
The core feature package can be extended by the MACsec feture package that includes:

- Sending and receiving of MACsec encrypted messages
- MACsec Key Agreement implementation

To get these features and suitable license files, contact: order@technica-engineering.de

4.5 Dashboard tab

This tab shows different counters of the switch. These include TX/RX counters for ethernet ports as well as counters for gPTP statistics. This tab also shows advanced counters regarding MACsec communication. The user can start/stop live updates of the counters as well as export the counters in *.csv format. It is possible to adapt the view/layout of the statistics and then export as well as import it as *.json file.



	BaseT1-1	BaseT1-2	BaseT1-3	BaseT1-4	BaseT1-5	BaseT1-6	BaseT1-7	BaseT1-8	SFP+ A	SFP+ B
Bytes Tx	0	0	0	0	0	0	0	0	0	0
Bytes Rx	0	0	0	0	0	0	0	0	0	0
Bad Bytes Rx	0	0	0	0	0	0	0	0	0	0
Bytes Tx in the Last Second	0	0	0	0	0	0	0	0	0	0
Bytes Rx in the Last Second	0	0	0	0	0	0	0	0	0	0
Bad Bytes Rx in the Last Second	0	0	0	0	0	0	0	0	0	0
Packets Tx	0	0	0	0	0	0	0	0	0	0
Unicast Packets Tx	0	0	0	0	0	0	0	0	0	0
Broadcast Packets Tx	0	0	0	0	0	0	0	0	0	0
Multicast Packets Tx	0	0	0	0	0	0	0	0	0	0
Packets Rx	0	0	0	0	0	0	0	0	0	0
Unicast Packets Rx	0	0	0	0	0	0	0	0	0	0
Broadcast Packets Rx	0	0	0	0	0	0	0	0	0	0
Multicast Packets Rx	0	0	0	0	0	0	0	0	0	0
Bad Packets Rx	0	0	0	0	0	0	0	0	0	0
Packets Tx in the Last Second	0	0	0	0	0	0	0	0	0	0
Packets Rx in the Last Second	0	0	0	0	0	0	0	0	0	0
Bad Packets Rx in the Last Second	0	0	0	0	0	0	0	0	0	0
Link Drop count	0	0	0	0	0	0	0	0	0	0
PTP Sync Rx	--	--	--	--	--	--	--	--	--	--
PTP FollowUp Rx	--	--	--	--	--	--	--	--	--	--
Valid PTP FollowUp Rx	--	--	--	--	--	--	--	--	--	--
Invalid Port Id PTP FollowUp Rx	--	--	--	--	--	--	--	--	--	--

Figure 4-40: Dashboard view

4.6 TC10 Events

TC10 stands for Technical Committee 10 of the OPEN Alliance, which defines the Sleep and Wake-up mechanisms for 100BASE-T1 and 1000BASE-T1 automotive Ethernet. TC10 enables energy-efficient Ethernet by allowing devices such as ECUs and switches to enter a low-power sleep state when idle and be remotely woken up via Ethernet signaling. The EES supports the following features:

- Trigger TC10 events: Sleep Request (LPS) and Wake-up Request (WUP).
- Read back PHY status.

The EES switch interface provides a dedicated TC10 Events view where users can trigger TC10 events, selecting between Sleep Request (LPS) and Wake-up Request (WUP) for each port. Note that the EES cannot awakened from sleep through TC10 wake-up events. For each port, acceptance of TC10 sleep requests can be individually configured. If TC10 Sleep Requests (LPS) are disabled on a port, the link partner cannot transition to the sleep state via TC10, and the link remains in the active (awake) state.

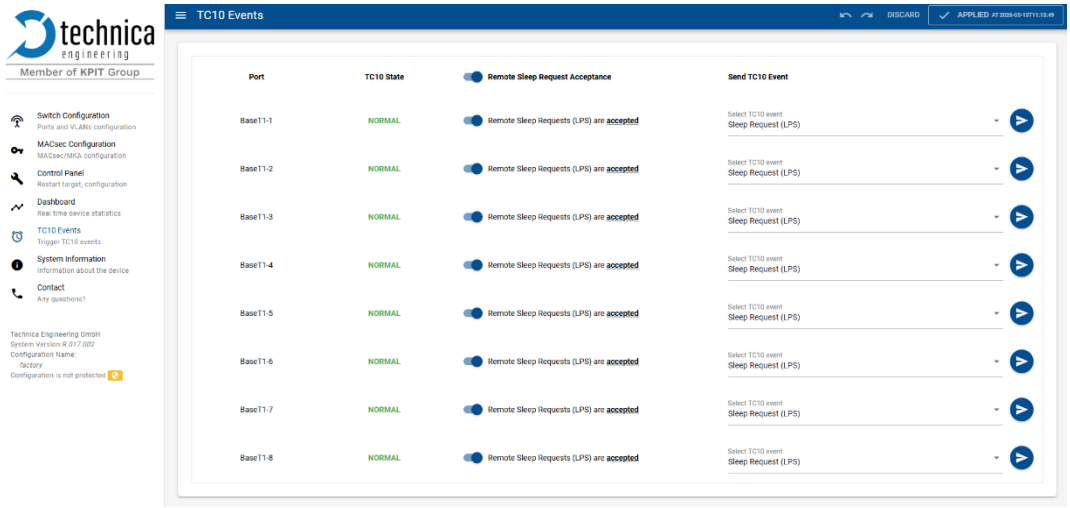


Figure 4-41: TC10 Events request.

Regarding the Monitoring/Read back of PHY status in real time which includes, states as Disabled, Normal, Sleep, Sleep Failed, and Sleep Aborted.

5 Firmware update

To perform the firmware update of the device it is necessary to download the current FW file from Customer portal of Technica Engineering: portal.technica-engineering.de/downloads

5.1 Preconditions and important information

To update the device to release 10 and onwards, the user must update the firmware to release 9 first and then update to release 10. A direct update from a lower release to release 10 can cause the device to break.

If an error occurs during the update, please perform a power reset, and try again

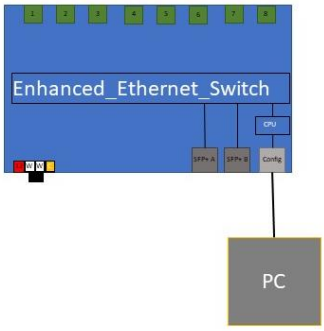
Make sure your network adapter is configured in the correct IP address range. The subnet mask should be 255.255.255.0

The device can only be flashed/updated through the 100BASE-TX Host port

Please follow these update instructions to avoid erroneous states of the device. Technica Engineering may charge support fees for repair services.

A stable 12 Volt DC power supply is required. Do not switch off the power supply during the update process.

5.1.1 Update via host port



Pre-requisites:

- Please set the device to default settings

Figure 5-1: Update through Host port

Steps:

1. Open the website through the host port and by typing the IP of the device. The Default IP address is 10.0.0.200
2. Go to the "System Information" tab and click System update

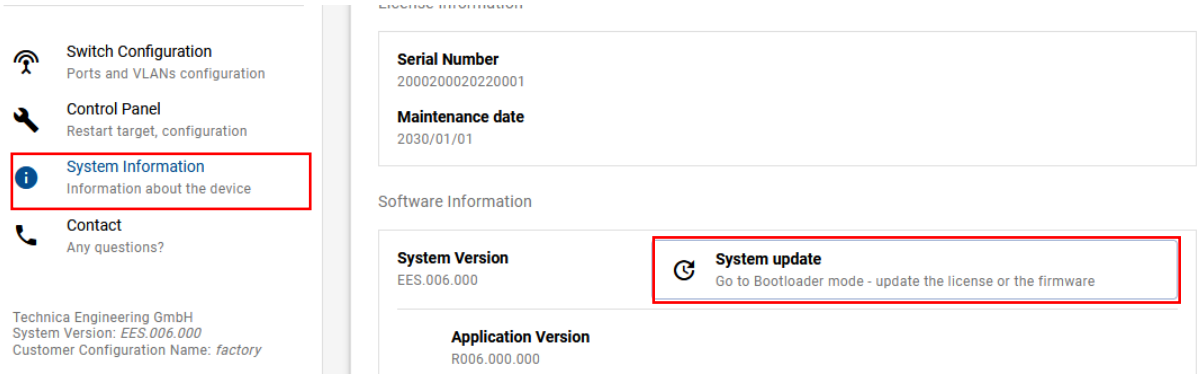


Figure 5-2: System information tab

The device is now in boot-loader mode, and the following website should appear:

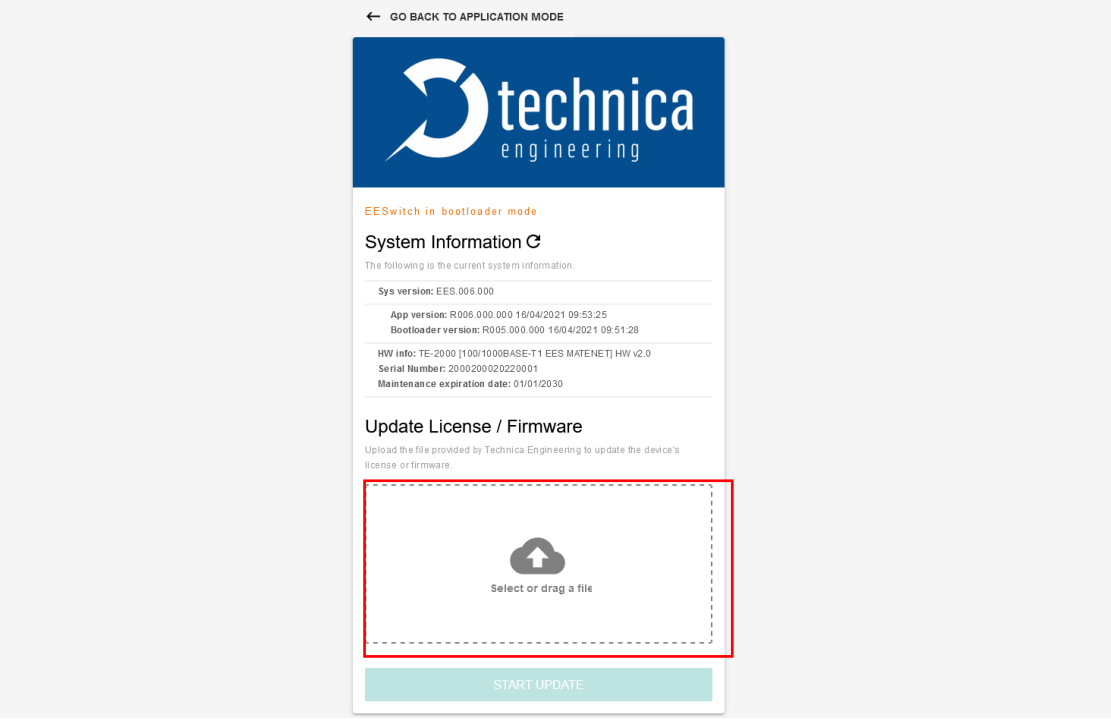


Figure 5-3: Firmware upload

3. Please click into the framed field and open the file *firmware_full.tefw* for new firmware or open new license files. Alternatively, it is possible to drag and drop the mentioned files into the framed field
4. Press *start update* START UPDATE
5. Wait until you are prompted by the following window:

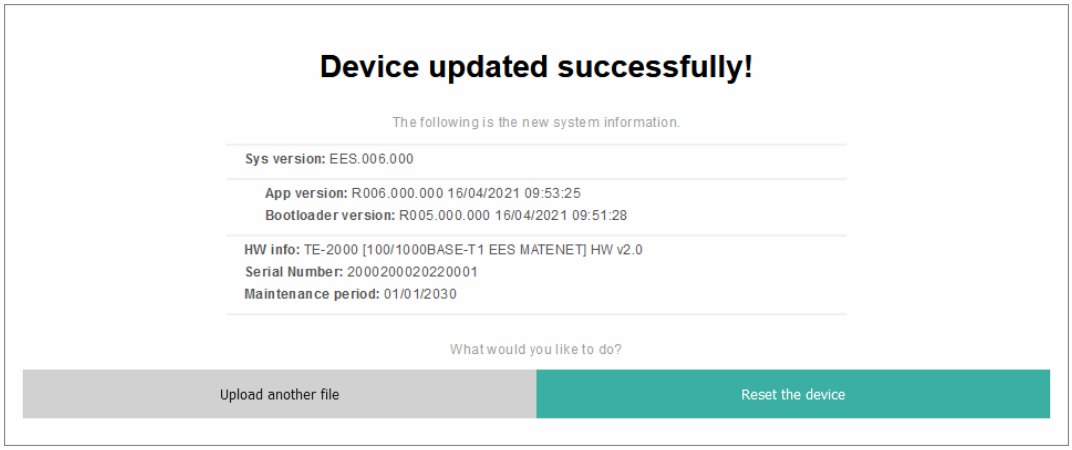


Figure 5-4: Firmware update successful

6. Press “Reset the device” if you are finished with updating.
7. The device will jump into application mode and show you the website again for configuration.

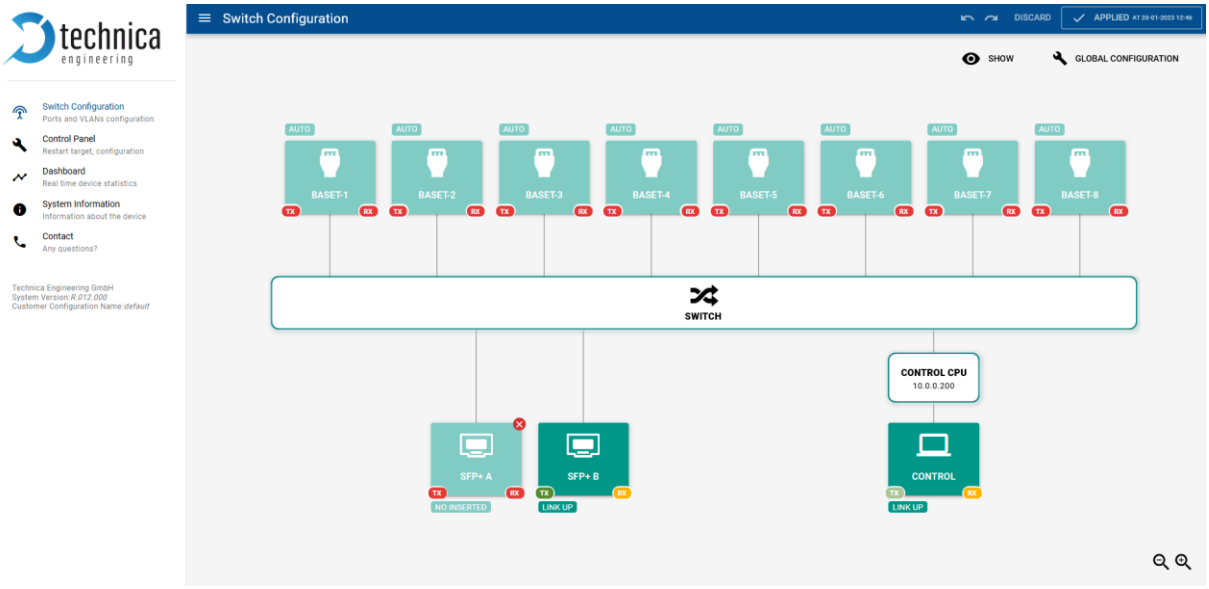


Figure 5-5: Home screen

If a newer version of a bootloader is present, then the same procedure has to be followed to update the bootloader.

6 Remote control API

The Enhanced Ethernet switch has a remote-control API based on HTTP protocol. Two types of methods are used for configuration.

- GET Methods: These allow retrieving the configuration and status using the .JSON format. They are used to read configuration and status information from the switch.
- Post Methods: These allow to setting and modification of the configuration coded and trigger actions

For detailed documentation of the remote API please contact Technica Engineering at Support.

7 Declaration of conformity/ Konformitätserklärung

English:

Hereby, Technica Engineering GmbH declares that the product Enhanced Ethernet Switch MACsec MATEnet (TE-2003) and Enhanced Ethernet Switch MACsec H-MTD (TE-2004) complies with Directive 2015/863/EU and 2014/30/EU. The full text of the EU declaration of conformity is available at the following internet address:

https://www.technica-engineering.de/

or on Customer portal:

https://portal.technica-engineering.de/downloads

Deutsch:

Hiermit erklärt Technica Engineering GmbH, dass das Produkt Enhanced Ethernet Switch MATEnet (TE-2003) und Enhanced Ethernet Switch H-MTD (TE-2004) die Richtlinie 2015/863/EU and 2014/30/EU entspricht. Der vollständige Text der EU-Konformitätserklärung ist unter der folgenden Internetadresse verfügbar:

https://www.technica-engineering.de/

oder im Kundenportal:

https://portal.technica-engineering.de/downloads

8 Important notes

8.1 General operating and safety strategy of Technica Engineering's Products

Technica Engineering's products are designed for operation in automotive systems and for supply voltage of typical 12 V (voltage range 6V - 30V). The applicable limit values adhere to the standard norms automotive onboard power systems correspondingly and can be found in the mentioned norms.

Should Technica Engineering's products be operated in voltage ranges beyond those specified in the norms, which represents a breach of the conditions of operation, then this will void the product warranty and Technica Engineering will assume no liability whatsoever for the results and/or consequences thereof.

This is especially valid whenever the voltage level reaches or exceeds the limits of the low-voltage directive. In this case, damage to the devices cannot be excluded. Due to the manufacturing characteristics of the devices, there is no imminent fire hazard from the device itself, if the devices are being operated in an environment according to the conditions of use. A secondary fire hazard cannot be excluded, should those conditions not be met. Protection against overvoltage cannot be provided in such a breach of the conditions of use.

8.2 General design rules for the power supply of Technica Engineering's products

The power supply solution of Technica Engineering's products is equipped with a self-protection mechanism. This automatic function protects the internal electronic components against excessive temperature and too high a supply voltage by switching the device off. This automatic switch-off function is independent of any software function. Excessive temperatures in the device can be caused by high environmental temperatures or internal failures of the device. The device can protect itself by shutting down its power supply avoiding further damage to itself.

The operator/user of these devices must take measures for the right protection of the power supply lines and the connector. The maximum power consumption of the Technica Engineering products is written in this user manual.

Please consider that some of the products work with a step-up converter. So, the current needed in case of undervoltage conditions must be considered higher than a typical voltage level. The user must choose the right fuse and/or other over current protection measures.

8.3 Warranty and Safety Information



Before operating the device, please read this User Manual thoroughly and follow the provided information and instructions. Keep the manual for future reference.



Operate the device only as instructed in this manual.
Ensure the device is used exclusively in dry conditions.
Refrain from inserting any foreign objects into the device's slots or openings.
Do not use or power a device that is damaged or malfunctioning.



Please do not attempt to open or disassemble the device. Unauthorized tampering or modifications will result in the immediate voiding of the warranty and may also compromise the safety and functionality of the device.



This product is intended for use in automobiles or similar environments, including office, laboratory, and workshop test setups. Technica Engineering products are not designed for standard IT use but as specialized test modules for experts in development and testing. Ensure adequate ventilation and proper external power supply with appropriate cable protection (e.g., fuses) when integrating these modules. Maximum current load specifications are detailed in the user manual. Technica Engineering products should be included in safety assessments and are aligned with standard Quality Management (QM) per ISO 26262.



Do not cover the device to prevent fire hazards.
Keep the device away from highly flammable materials to avoid fire risk.
Do not operate the device above the specified temperature limits.
Note that the operating temperature refers to the ambient temperature of the installation area.



This symbol is applicable only within the European Union. To discard this product, please consult your local authorities or dealer for proper disposal methods. Technica Engineering GmbH is registered as a manufacturer of the "Technica Engineering" brand, specializing in "Small devices of Information and Telecommunications Technology for exclusive use in non-private households." WEEE Registration Number: DE 20776859.



This product has been manufactured and tested to comply with CE (Conformité Européenne) standards, as indicated by the EU Declaration of Conformity.

9 Wichtige Hinweise

9.1 Allgemeine Betriebs- und Sicherheitsstrategie für Produkte von Technica Engineering

Die Produkte von Technica Engineering sind für den Einsatz in Fahrzeugsystemen und für eine typische Versorgungsspannung von 12 V (Spannungsbereich 6V - 30V) ausgelegt. Die geltenden Grenzwerte entsprechen den Normen der Bordnetze von Kraftfahrzeugen und sind in den genannten Normen zu finden.

Sollten die Produkte von Technica Engineering in Spannungsbereichen betrieben werden, die außerhalb der in den Normen angegebenen Bereichen liegen, was einen Verstoß gegen die Betriebsbedingungen darstellt, erlischt die Produktgarantie und Technica Engineering übernimmt keinerlei Haftung für die daraus resultierenden Ergebnisse und/oder Folgen.

Dies gilt insbesondere dann, wenn das Spannungsniveau die Grenzwerte der Niederspannungsrichtlinie erreicht oder überschreitet. In diesem Fall kann ein Schaden an den Geräten nicht ausgeschlossen werden. Aufgrund der Herstellungseigenschaften der Geräte besteht keine unmittelbare Brandgefahr durch das Gerät selbst, sofern die Geräte in einer Umgebung gemäß den Nutzungsbedingungen betrieben werden. Eine sekundäre Brandgefahr kann jedoch nicht ausgeschlossen werden, falls diese Bedingungen nicht eingehalten werden. Ein Schutz gegen Überspannung kann, in einem solchen Fall von Verstoß gegen die Nutzungsbedingungen, nicht gewährleistet werden.

9.2 Allgemeine Designregeln für die Stromversorgung von Produkten von Technica Engineering

Die Stromversorgungslösung der Produkte von Technica Engineering ist mit einem Selbstschutzmechanismus ausgestattet. Diese automatische Funktion schützt die internen elektronischen Komponenten vor übermäßiger Temperatur und zu hoher Versorgungsspannung, indem das Gerät abgeschaltet wird. Diese automatische Abschaltfunktion ist unabhängig von jeglichen Softwarefunktionen. Übermäßige Temperaturen im Gerät können durch hohe Umgebungstemperaturen oder interne Fehler des Geräts verursacht werden. Das Gerät kann sich durch Abschalten seiner Stromversorgung selbst schützen und so weiteren Schaden vermeiden.

Der Betreiber/Nutzer dieser Geräte muss Maßnahmen ergreifen, um den richtigen Schutz der Stromversorgungsleitungen und des Steckverbinders sicherzustellen. Der maximale Stromverbrauch der Produkte von Technica Engineering ist in diesem Benutzerhandbuch angegeben.

Bitte beachten Sie, dass einige der Produkte mit einem Step-Up-Konverter arbeiten. Daher muss berücksichtigt werden, dass der Strombedarf im Falle von Unterspannung höher sein kann als bei einem typischen Spannungsniveau. Der Nutzer muss die richtige Sicherung und/oder andere Überstromschutzmaßnahmen wählen.

9.3 Garantie- und Sicherheitsinformationen



Bevor Sie das Gerät in Betrieb nehmen, lesen Sie bitte dieses Benutzerhandbuch sorgfältig durch und befolgen Sie die angegebenen Informationen und Anweisungen. Bewahren Sie das Handbuch als zukünftige Referenz auf.



Betreiben Sie das Gerät nur gemäß den Anweisungen in diesem Handbuch.
Stellen Sie sicher, dass das Gerät ausschließlich unter trockenen Bedingungen verwendet wird.
Vermeiden Sie es, Fremdkörper in die Schlitze oder Öffnungen des Geräts einzuführen.
Verwenden oder betreiben Sie kein Gerät, das beschädigt oder defekt ist.



Bitte versuchen Sie nicht, das Gerät zu öffnen oder auseinanderzubauen. Unbefugte Manipulationen oder Modifikationen führen zum sofortigen Erlöschen der Garantie und können zudem die Sicherheit und Funktionalität des Geräts beeinträchtigen.



Dieses Produkt ist für den Einsatz in Automobilen oder ähnlichen Umgebungen bestimmt, einschließlich Büro-, Labor- und Werkstatt-Testaufbauten. Produkte von Technica Engineering sind nicht für den Standard-IT-Einsatz konzipiert, sondern als spezialisierte Testmodule für Experten in der Entwicklung und Prüfung. Sorgen Sie für eine ausreichende Belüftung und eine ordnungsgemäße externe Stromversorgung mit geeigneter Kabelsicherung (z.B. Sicherungen) beim Einbau dieser Module. Die maximalen Stromlastspezifikationen sind im Benutzerhandbuch detailliert beschrieben. Produkte von Technica Engineering sollten in Sicherheitsbewertungen einbezogen werden und sind mit dem Standard-Qualitätsmanagement (QM) gemäß ISO 26262 abgestimmt.



Bedecken Sie das Gerät nicht, um Brandgefahren zu vermeiden.
Halten Sie das Gerät von leicht entflammbaren Materialien fern, um Brandrisiken zu vermeiden.
Betreiben Sie das Gerät nicht außerhalb der angegebenen Temperaturgrenzen.
Beachten Sie, dass sich die Betriebstemperatur auf die Umgebungstemperatur des Installationsbereichs bezieht.



Dieses Symbol ist nur innerhalb der Europäischen Union gültig. Um dieses Produkt zu entsorgen, konsultieren Sie bitte Ihre örtlichen Behörden oder Ihren Händler für eine ordnungsgemäße Entsorgungsmethode. Technica Engineering GmbH ist als Hersteller der Marke „Technica Engineering“ registriert und spezialisiert auf „Kleingeräte der Informations- und Telekommunikationstechnologie zur ausschließlichen Nutzung in nicht-privaten Haushalten“. WEEE-Registrierungsnummer: DE 20776859.



Dieses Produkt wurde hergestellt und getestet, um den CE-Standards (Conformité Européenne) zu entsprechen, wie durch die EU-Konformitätserklärung angegeben.

9.4 IMPRINT

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Website: www.technica-engineering.de

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